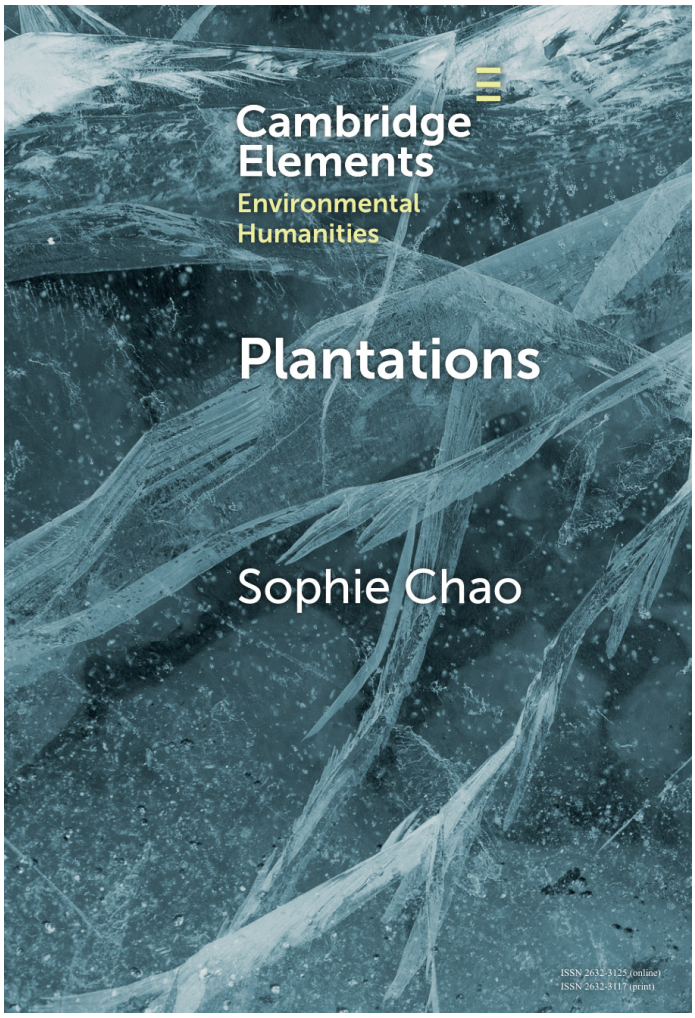
Plantations: Extraction, Extinction, Emergence

Sophie Chao

English reading edition

Complete reading edition based on the local Cambridge University Press PDF (2026), including all five sections, 276 references, the cover, and supplementary pages. Margin labels use original printed pages (p.) or PDF sequence numbers for unpaginated matter. Paragraphs continuing across source pages are joined. Source wording and bibliographic anomalies are retained.



Plantations Sophie Chao Environmental Humanities ISSN 2632-3125 (online) ISSN 2632-3117 (print)

Elements in Environmental Humanities

edited by Serenella Iovino *University of North Carolina at Chapel Hill* Timo Maran *University of Tartu* Thom van Dooren *The University of Sydney*

PDF 3

PLANTATIONS Extraction, Extinction, Emergence

Sophie Chao *University of Sydney*

Shaftesbury Road, Cambridge CB2 8EA, United Kingdom One Liberty Plaza, 20th Floor, New York, NY 10006, USA 477 Williamstown Road, Port Melbourne, VIC 3207, Australia 314–321, 3rd Floor, Plot 3, Splendor Forum, Jasola District Centre, New Delhi –110025, India

PDF 4

Cambridge University Press is part of Cambridge University Press & Assessment, a department of the University of Cambridge. We share the University's mission to contribute to society through the pursuit of education, learning and research at the highest international levels of excellence.

www.cambridge.org Information on this title: www.cambridge.org/9781009531252 DOI: 10.1017/9781009531245

© Sophie Chao 2026 This publication is in copyright. Subject to statutory exception and to the provisions of relevant collective licensing agreements, no reproduction of any part may take place without the written permission of Cambridge University Press & Assessment.

When citing this work, please include a reference to the DOI 10.1017/9781009531245 First published 2026

A catalogue record for this publication is available from the British Library A Cataloging-in-Publication data record for this Element is available from the Library of Congress

ISBN 978-1-009-53125-2 Hardback ISBN 978-1-009-53122-1 Paperback ISSN 2632-3125 (online) ISSN 2632-3117 (print)

Cambridge University Press & Assessment has no responsibility for the persistence or accuracy of URLs for external or third-party internet websites referred to in this publication and does not guarantee that any content on such websites is, or will remain, accurate or appropriate.

For EU product safety concerns, contact us at Calle de José Abascal, 56, 1°, 28003 Madrid, Spain, or email eugpsr @cambridge.org

Plantations Extraction, Extinction, Emergence

PDF 5 Elements in Environmental Humanities DOI: 10.1017/9781009531245 First published online: June 2026 Sophie Chao *University of Sydney* Author for correspondence: Sophie Chao, sophie.chao@sydney.edu.au

Abstract: Plantations are major drivers of biodiversity loss, habitat degradation, and climate change. They find root in (neo)colonial logics of mastery and progress that position nature as a passive resource, exploited to serve (certain) humans' ends. Yet the rise and fall of plantations have never been determined entirely by those humans and institutions who claim to create and control them. Rather, plantations are animated by entangled processes of multispecies extraction, extinction, and emergence. This Element considers the violence and vulnerabilities engendered by plantations for differently positioned humans and non-humans ⁻from indentured labourers, displaced communities, and environmental activists, to soils, parasites, and crops. It examines how acts of resistance, alliance, and solidarity have challenged the dominance of plantations over places, plants, and peoples. Approaching plantations as fertile sites for theorizing inter- and intra-human relations, the Element unearths in their troubled terrains unexpected yet urgent possibilities for cultivating counter-plantation futures and multispecies justice.

This Element also has a video abstract: www.cambridge.org/EIEH_Chao_abstract **Keywords:** plantations, colonialism, capitalism, violence, justice

© Sophie Chao 2026 ISBNs: 9781009531252 (HB), 9781009531221 (PB), 9781009531245 (OC) ISSNs: 2632-3125 (online), 2632-3117 (print)

Contents

PDF 6 1 Introduction 1

2 Extraction 9

3 Extinction 24

4 Emergence 39

5 Conclusion 55

References 64

1 Introduction

There is a story in rural West Papua that tells of the cassowary, *khei*, the founding spirit of the living world. At once creator, brother, father, and progeny, the cassowary brought forth humans, animals, and plants from a primordial mud, beaten and thrashed into form by the towering bird's webbed feet over the span of a hundred thousand moons. It was the cassowary's beak that nudged our knees and elbows into shape. It was the cassowary's wattle that tickled our ears and noses into being. It was the cassowary's feathers that slickened and smoothed our surfaces into skin. It was the cassowary's grunt that tuned and tempered the pitch of what was later to become our voice. It is to the cassowary that we owe these traits –and it is to the cassowary that they return upon our passing. p. 1

My first encounter with a cassowary took place along a dusty highway during long-term ethnographic fieldwork I was conducting among Indigenous Marind Peoples in the Indonesian-occupied region of West Papua –a place where the historical theft of sovereignty over lands, livelihoods, and bodies has, in recent years, been amplified by the proliferation of large-scale industrial oil palm plantations. I had, at the time, heard many stories from villagers about this mythical, revered, sacred, and mysterious being of solitary tendencies, whose presence is often known less through direct encounters than through the traces it leaves upon the landscape –deep footprints etched in the mud, sporadic rumbles overheard in the sago grove, wispy or wiry feathers caught on overhanging vines and lianas, and crushed berries strewn on the forest floor, bearing the characteristic dents of the bird's powerful beak.

But this cassowary was different. Noble and majestic it certainly was, with its sleek, iridescent plumage, and strong muscular legs. But it was also confused, distraught, terrified. The bird was standing in the middle of a dusty plantation road, caught by surprise as trucks, cars, and jeeps rushed past. It looked with panicked eyes to the licking flames of a burning forest on the left –one of many victims to the illegal land clearing through burning taking place at this time of the year. Then, it turned its gaze to the vegetal barricade of thorny oil palm trees on the right – a newly established plantation of over 50,000 hectares that now stood where a sacred sago grove had once flourished. Marinus, an Indigenous land rights activist and fellow traveller, turned to me and said, 'See, the *khei*'s fate is just like ours. The *khei* has lost its home, it has no kin to flee to, its ancestors are turning into dust because of oil palm. The *khei* inhabits a strange and dangerous place. Like us, it is confused and afraid. Like us, it is struggling to survive'. No sooner had my friend uttered these words than the cassowary dove with a desperate screech into the blazing forest, seemingly choosing flame over famine.

And so, I wondered in this moment: what kinds of justice can exist in a place where humans, non-humans, and ancestors find themselves caught in the colliding ecologies of forest and plantation? Who gets to story the lives and deaths of beings alternately sacrificed or starved at the pyre of agro-industrial capitalism, in all their unspeakable violence and vulnerability? What are the social and environmental consequences of a monocultural production system that is premised upon and perpetuates multispecies erasure and biocultural erosion? What possibilities might exist for more-than-human coexistence and collective future-building amidst the disfigured landscapes of the plantation zone? p. 2

1.1 Plantations as More-than-Human Landscapes

The plantation landscape described in this opening vignette constitutes one patch within what some environmental humanities scholars call the 'Plantationocene' –a spatiotemporal formation that entails 'the devastating transformation of diverse kinds of human-tended farms, pastures, and forests into extractive and enclosed plantations, relying on slave labour and other forms of exploited, alienated, and usually spatially transported labour' (Haraway 2015: 162). Today, monocultures cover some 80 per cent of the world's arable land. As of 2019, oil palm plantations occupied 12 per cent of the land cover of Indonesia –the world's top palm oil-producing country –and over 19 million hectares globally, with the figure quadrupling in the last fifty years (see Figure 1). Industrial soy –the fastest-growing cash crop in the world – now extends across 125 million hectares, with production doubling in the last twenty years. The proliferation of these and other so-called 'flex crops', like sugarcane and jatropha, has more recently been boosted by their promotion as sources of renewable biofuel, bioelectricity, and bioplastic, alongside food for humans and livestock. The international food, fuel, and finance crisis of 2008 has been a major accelerator of the large-scale acquisition and transformation of land for plantation developments across the Global South (Borras & Franco 2011).

Monocrop plantations as such are emblematic of an era in which industrial activities are reconfiguring landscapes at a planetary scale in the name of profit, productivity, and progress. Within these monocultured spaces, variously described as 'open-air factories' (Lévi-Strauss 1961: 99) and exemplars of 'military agriculture' (Thompson 1975), visions of out-and-out exterminism and regimented homogeneity work hand in hand with dreams of endless growth. The consequences of these visions include damaged native ecosystems, soil degradation, freshwater depletion, chemical contamination, and exacerbated climate change and global warming resulting from rising greenhouse gas emissions. Plantations have also furthered the dispossession of Indigenous Peoples like the Marind of West Papua, as well as many other rural communities across the tropics, whose intimate and ancestral relations to land are routinely ob- p. 2–3

scured, neglected, or violated by the state and corporate actors promoting and implementing monocrop developments (Chao 2022a, 2025). In both their human and ecological impacts, plantations as industrial infrastructures are indelibly tethered to longer histories of colonial expansion, capitalist incursion, territorial appropriation, corporate occupation, exploitative labour, and entrenched nature-culture divides that together legitimate the unsustainable extraction of natural resources and the subjection of marginalized humans to racialized hierarchies of worth (Koshy et al. 2022).



Figure 1 Aerial view of an oil palm plantation and adjacent road. Image credit: Pok Rie.

p. 3–4 And yet, the rise and fall of plantations as 'capitalist natures' (Escobar 1999: 6, 11) have never been determined entirely by the humans and institutions who claim to create and control them. Nor are their forms and effects limited in substance and scope to the confines of monocrops as physical infrastructures, let alone distinguishable along rigid human-nonhuman divides. Rather, as this Element uncovers, plantations have always been and continue to be shaped by entangled processes of *multispecies* extraction, extinction, and emergence. Extraction centres how commodities come into being through the harnessing of humans' and plants' bodies, labours, metabolisms, and affordances as exploited, uprooted, and often un(der)acknowledged workers. Extinction draws attention to more-than-human losses, erasures, and erosions of ways of being and relating as the by-products of industrial plantation production regimes. Emergence focuses on the new relations, institutions, and practices generated by monocrop systems, including those that challenge plantations' technocapitalist, colonial, and racializing logic. This tripartite approach serves to structure the material that follows, but in a non-deterministic or mutually exclusive manner. For instance, extraction, extinction, and emergence can – and often do – occur simultaneously rather than sequentially. Similarly, what is experienced as extinction by one party might be apprehended as emergence by another. Human and other-than-human interests may align in certain instances, and at others diverge – at different moments, in different contexts, and to different ends. Importantly, the approach adopted in this work does not mark a shift away from the impacts of plantation logics on human communities – from questions of labour, political economy, and power, to realities of exploitation, violence, and vulnerability, among others. Rather, it invites a transformative expansion, pluralization, and cross-pollination of the ways in which these recurring motifs can be examined through an attention to their human *and* other-than-human manifestations, intersections, and frictions.

p. 4 In approaching plantations as 'human but not only' (de la Cadena 2014), this Element thus reads against the grain of fictive yet enduring nature-culture binaries while pointing to the multifarious consequences of these divides for variously positioned human and other-than-human entities. It attends to how plantations make, unmake, and remake multispecies worlds as they enlist diverse protagonists in the production of new configurations of life and death, and of order and disorder. In doing so, the Element seeks to (re)tell plantation stories in ways that invite, provoke, reflect upon, and challenge assumptions of human exceptionalism, and that remain attentive to the complex intersections of race, coloniality, capitalism, and species animating plantations as 'landscapes of empire' (Besky 2013: 3).

1.2 Positionalities and Audiences

I write this Element as an anthropologist and former human rights advocate who has conducted over a decade of ethnographic and investigative fieldwork in the Southeast Asian oil palm sector. Through my academic and activist career, I have sought to understand how Indigenous communities experience, theorize, and contest the adverse repercussions of plantation proliferation on their ways of knowing, being, and relating to human and non-human lifeforms. It is over the course of long-term, immersive participant-observation among Indigenous communities in the Indonesian-occupied region of West Papua that I became attuned to the multispecies dimensions of plantationscapes, which I grapple with from a trans-geographic and trans-temporal perspective in this Element. It is also this field-based research that has brought me to reflect on broader questions that haunt and heighten the ethical, political, and intellectual stakes of plantation studies. Who are the subjects of justice in plantation landscapes as more-than-human contact zones? What insights arise from thinking about plantation existence and counter-plantation resistance through a multispecies lens? In an age of ecological unmaking, what can plantation lifeworlds and afterlives teach us about the more-than-human dimensions of what Judith Butler (2004: 48) calls our 'common corporeal vulnerability'? p. 4–5

In engaging with these questions, this Element responds to calls stemming from the environmental humanities to foster ways of knowing, engaging, and storytelling that recognize the significant lives of non-human others and enable us to cultivate other kinds of connections, accountabilities, and obligations towards the more-than-human world (van Dooren & Rose 2016: 77, 89). This approach articulates a 'thicker notion of humanity' that centres the embeddedness ⁻rather than hyperseparation ⁻of humans *within* multispecies communities (van Dooren & Rose 2012: 1; see also Plumwood 2002). It draws attention to ways of knowing more-than-human worlds that fall beyond the ambit of dominant, Western-rooted and secular scientific epistemologies, and that have been created, fashioned, and adapted to changing socio-ecological contexts by Indigenous Peoples, as well as by enslaved and otherwise exploited and marginalized communities. p. 5

More specifically, this Element intervenes into environmental humanities debates on the origins and effects of the plantation form, which have seen renewed resurgence in recent years. Critical engagements with the concept of the 'Plantationocene' in particular have revitalized conversations across the social sciences, arts, and humanities on the relation between human and other-than-human violence and vulnerability within plantations (Davis et al. 2019; Jegathesan 2021; Chao, Wolford, Ofstehage et al. 2024). Transdisciplinary, multi-sited, and comparative approaches, such as the one adopted in this Element, are increasingly seen as necessary in understanding and analysing monocrop systems and societies (Forney, Rosin & Campbell 2018). Such comparative work, as I hope to demonstrate in the material that follows, helps uncover recurring motifs that have accompanied plantation proliferation globally, but also sheds vital light on plantations' multiple and situated manifestations across space and time. It reveals how plantation imaginaries of simplification, scalability, and mastery over people, plants, and ecosystems in fact often operate alongside realities of plantation failure, patchiness, and contingency that belie their apparent homogeneity and timelessness. It uncovers how some plantations embody more explicitly than others the dynamics of extraction, extinction, and emergence ⁻and how each plantation expresses these dynamics in its own particular way. It also invites us to reconsider all agro-industrial formations and legacies through their physical, ideological, and historical similarities and differences to those systems that are commonly labelled and recognized as 'plantations' per se. p. 5–6

On one plane, this Element is written for students and scholars in the environmental humanities and consonant disciplines, including anthropology, political ecology, multispecies studies, environmental history, geography, critical race studies, and science and technology studies. But it is equally of relevance to policymakers, non-governmental organizations, journalists, industry actors, and indeed, any member of the informed public who is interested in understanding plantations' origins and forms and who is concerned or affected by plantations' social and environmental impacts. On another plane, the Element speaks to all of us humans who inhabit this increasingly wounded and wretched place we call Earth (Gray & Sheikh 2018; Chao 2023). For wherever we happen to be rooted in the world, and no matter whether we have ever set foot in a monocrop, the foods and other resources that are produced through plantation systems are an intrinsic part of our everyday consumer lives. Raising awareness of our often invisibilized connections to, and complicities with, plantation violence is part of what this Element seeks to achieve, alongside identifying possibilities for creating less extractive relations to the plants, places, and peoples who fuel and feed us. p. 6

In the pages that follow, readers will travel diverse world-regions and time-periods through ethnographic and historical case studies that exemplify how plantations have been shaped by the agencies of both human and other-than-human protagonists ⁻from indentured labourers, displaced communities, and environmental activists, to soils, parasites, and cash crops themselves. We will encounter, among many other bio-geographies, the colonial sugarcane, cotton, tobacco, hemp, and rubber estates of the Caribbean, United States, Hawai'i, Brazil, Argentina, and Liberia. We will journey through contemporary oil palm, soy, tea, timber, and banana plantations in Indonesia, Cambodia, Italy, India, Paraguay, and the Philippines. We will consider these *literal* fields through the lens of multiple *scholarly* fields ⁻from anthropology, critical race theory, geography, and history to political ecology, literary studies, agronomy, and plant biology. In the process, the diverse ways in which plantations are produced and experienced across space and time

come to mirror the diverse ways in which they have been understood and addressed across disciplines.

1.3 Section Outline

p. 7 In Section 1, I consider how *extraction* in the plantation sector manifests in the coerced displacement of Indigenous Peoples from lands targeted for monocrop development under colonial regimes, in the uprooting and transplanting of enslaved and indentured workers from continents far removed from plantation geographies, and in the extraction of labour from the bodies of people subject to plantations' racializing regimes. I also attend to how extraction shapes the worlds of cash crops as sources of metabolic work along with the worlds of so-called beneficial species whose biotic affordances are harnessed by corporations to sustain plantations' ecological health under biological pest management programmes. In tandem, I reflect on instances where corporate-colonial systems of extraction fail, face resistance, or are otherwise threatened by the unexpected actions of human and non-human lifeforms – a topic I return to in further detail in Section 3.

In Section 2, I examine plantation lifeworlds and afterlives as drivers of multispecies *extinctions* that are at once local and global, ecological and sociocultural. I trace the accelerating loss and critical endangerment of native flora and fauna that often accompany the substitution of biodiverse forests with industrial monocrops, situating this dynamic within a broader epoch of ecosystemic decline and planetary crisis. I uncover how the apparent homogeneity of monocrops masks realities of complex parasitic and mutualistic entanglements – some of which sustain cash crop yield and health, and others of which undermine it. I also consider the erosion of multispecies worlds and biocultural lifeways among communities whose lands are transformed into privatized plantations, reshaping in the process a range of more-than-human socialities, relationalities, and identities. I conclude by reflecting on the promise and perils of sustainability paradigms and conservation schemes that aim to offset the ecological harms caused by plantations, yet that often reinforce exclusion and dispossession through top-down approaches to environmental preservation and (neo)colonial framings of 'nature' as pristine wilderness.

p. 7–8 In Section 3, I turn my attention to modes of human and non-human being and becoming that find root in, yet meaningfully push against, the putative dominance and mastery of monocrop regimes over instrumentalized landscapes and lifeforms. I consider these modes of *emergence* in the context of plantations as laboratories for the production of colonial, postcolonial, and anti-colonial epistemologies of race, disease, and science, highlighting the frictions and cross-pollinations across different understandings of illness and practices of healing. I examine how plantations generate diverse meanings, moralities, and materialities through people's engagements with cash crops, producing new vegetal-human intimacies while also shaping contemporary foodways and fuelling anti-colonial and anti-capitalist food sovereignty movements. I further consider how the racializing and ecocidal logics of plantations have given rise to new and counter-hegemonic identities, imaginaries, aspirations, and practices including art and aesthetics as politically charged sites of expression wherein plantations' destructive yet contested effects are creatively represented, critiqued, resisted, and reimagined.

p. 8 In the Conclusion, I highlight the plurality of plantation systems and societies and foreground the urgency and challenge of attending to plantations through their concomitant inter- and intra-species dynamics. I consider the various ways in which plantation logics endure beyond physical agricultural fields and reflect on the promise of (counter)plantation futures and multispecies justice that might be unearthed in the troubled terrains of monocrop lifeworlds and their enduring afterlives.

1.4 Terminology

Finally – a few terminological clarifications. First, I intentionally refer in this Element to 'plantations' in the plural rather than to 'plantation' in the singular. I do this to underscore plantations' situated and shapeshifting manifestations across space and time as recursive formations, rather than as homogeneous or monolithic systems in either existence or endurance (López 2011: 402; Thomas 2023: 354). Indeed, one of the key aims of this Element is to uncover how ultimately, and despite some prominent and recurring motifs, there is no such thing as 'the' plantation in any singular or static sense, but rather a *plantation multiple* that varies across spatio-temporally defined contexts, practices, ideologies, communities, and the particular constellation of relations that connect and shape them.

p. 8–9 Second, while I use the terms 'plantations' and 'monocrops' interchangeably within this Element, my primary focus lies more specifically with what Miles Kenney-Lazar and Noboru Ishikawa (2019: 63) term 'mega-plantations', or 'plantation development that rapidly and radically transforms landscapes in ways that displace and replace preexisting human and nonhuman communities' and that 'require the application of large amounts of capital and political power and the transnational organization of labour, capital, and material'. Even within this definitional scope, I do not presume to cover plantation systems in all their taxonomic, temporal, social, economic, ecological or geographic diversity. To claim such a comprehensive and all-encompassing grasp on the myriad dimensions of plantation lifeworlds would be

to problematically replicate assumptions of universalism and epistemic authority that are central to the ideology and operation of plantation logic itself.

Third, I use the term ‘multispecies’ in this Element to refer to lifeforms of all kinds, including plants, animals, and humans, but also elements, atmospheres, and landscapes that sustain human and non-human existence (Price & Chao 2022). This expansive definition makes space for abiotic entities that do not fit comfortably within Western secular scientific understandings or valuations of ‘life’, yet that have always been deemed sentient, animate, or otherwise agentive in many non-Western worldviews, including soils, waters, spirits, and the deceased (TallBear 2015; Chao 2026). It also positions humans within, rather than outside, the scope of ‘multispecies’, thereby allowing me to approach plantations as fertile –if fraught –grounds for engaging with the conjoined poetics and politics of race, culture, and nature (Moore, Pandian & Kosek 2003), as these shape and are shaped by, differential gradations, hierarchies, and categories of worth –within, across, and beyond species lines.

p. 9

2 Extraction

p. 9–10 The root of the English word ‘plantation’ can be traced back to fourteenth-century feudal Europe, when the term was used to refer to vanguard colonial settlements, including towns, farms, and seigniories, and particularly to the establishment of settler communities and industries in overseas colonies (Moore 2010; Manjapra 2018). Natural resource extraction, alongside human displacement, imperial expansion, and territorial occupation, were thus key operatives within plantation semantics and systems from the outset. In many instances, these dynamics sit within and further dominant visions of extraction as a ‘civilizational’ process that transforms putatively ‘wild’ landscapes, together with their human and non-human dwellers, into disciplined or tamed subjects whose bodies and affordances are put to work to serve colonial and postcolonial agendas. Historically, the conjuring of extraction as a form of progress and improvement alongside a source of profit and productivity, manifested in the transformation of jungle-laden frontiers into monocrop tea plantations in imperial India and of swamp ecologies into sugarcane plantations in the colonial Caribbean. Today, this logic continues to legitimize the large-scale substitution of tropical forests for timber monocrops in Indonesia, of biodiverse marshlands for oil palm plantations in Colombia, and of subtropical grasslands for soy agro-industries in Argentina. In examining how extraction ⁻a process often associated primarily with subterranean industrial activities such as mining ⁻shapes plantation lifeworlds and afterlives, I invite a return to the term’s original and more expansive sense of a literal or figurative, and often coercive, *drawing out*, *withdrawing*, or *removal* in space and time that reshapes the existences and entanglements of humans and other-than-humans. Doing so uncovers extraction’s complex relationship with value and worth, as these are at times fiercely coveted and guarded, and at others discarded and invisibilized.

2.1 Indigenous Dispossession and Rural Displacement

p. 10 The establishment of monocrops in (settler)-colonized world-regions was from the very outset enabled through the forceful extraction of Indigenous Peoples from lands with which they entertained intimate cultural, spiritual, and ancestral relations since time immemorial. In the early nineteenth-century American South, for example, the spread of cotton plantations, alongside tobacco and hemp industries, was facilitated by the expulsion of Native Peoples, including the Cherokee, Chickasaw, Creek, and Seminole from their homelands in Georgia, Alabama, Tennessee, and North Carolina, to new territories (or reservations) west of the Mississippi River, in a process that was legally instituted and sanctioned through the Indian Removal Act of 1830. Over one-quarter of the Cherokee population alone died of cold, hunger, and disease during the relocation journey of 1838⁻1839, which was later designated and commemorated as the ‘Trail of Tears’ through a 1987 Act of Congress (Perdue 2012; Kennedy 2013). The imposed extraction of Indigenous Peoples from their native lands was also seminal to the establishment of sugarcane plantations in seventeenth-century Dutch-colonized Guyana, where imperial agribusiness ventures forced communities who had inhabited the Caribbean islands for some 6,000 years to retreat from the Atlantic coast to the hinterland or interior ⁻a zone that continues to be imagined and experienced as a ‘space of underdevelopment’ (Jackson 2012) compared to areas earlier integrated into the imperial political economy. In the teeth of colonial extraction, the cultural, economic, and political power of Indigenous Peoples across the broader Caribbean region was rendered secondary to those of settler-Creoles, creating new and often-fraught identity struggles between Black and Native communities that perdure to this day.

p. 10–11 Further south in Latin America, sugar plantations developed under sixteenth and seventeenth-century Portuguese rule were largely enabled through the systematic uprooting of various Indigenous communities, notably speakers of Tupi⁻Guarani languages in Northeast Brazil, some of whom were initially incorporated as labourers within sugarcane concessions but were then replaced with imported and enslaved Africans. By 1590, malnourishment, exhaustion, and diseases including measles and haemorrhagic fever, had reduced Indigenous populations to a tenth of their original numbers. The ‘structures of violence and death’ (Eichen 2020: 36) undergirding plantation proliferation thus encompassed both the conversion of Indigenous Peoples’ biodiverse native forests into imperial sources of wealth *and* the extraction of Indigenous Peoples themselves from the native soils, landscapes, and ecosystems that once sustained and nourished them.

p. 11 Importantly, plantation expansion across these and other colonial periods and geographies did not operate in a silo but rather sat within a larger nexus of imperial projects seeking to assert physical and political control over colonies as natural resources and over colonized subjects as targets of displacement, eradication, and/or exploitation ⁻from mining, urbanization, militarization, and road and other infrastructural developments to enforced missionization, formal education, and cultural assimilation. Today, many plantations in the Global South remain predicated on extraction and occupation as the axiomatic premises for the enhancement of productivity and profit, albeit often framed in the language of globalization, modernization, national economic growth, and regional and rural poverty alleviation.

In present-day Cambodia, for instance, up to 800,000 people have suffered from land conflicts, livelihood losses, and forced displacement following the establishment of Economic Land Concessions (ELCs), a long-term lease mechanism that allows beneficiaries to clear land in order to develop industrial agriculture and that was revived by the

Cambodian government in the 1990s to enhance foreign investment and economic development. Related government policies that encourage populations displaced by ELCs to resettle in peripheral territories that are classified as 'unused' or 'unoccupied' under state legislation have prompted intensifying conflict between incoming migrants and local communities (Han & McAndrew 2018). ELCs have had particularly marked impacts among rural ethnic minorities in Cambodia, many of whom depend on the natural environment for their income and subsistence, including through rice paddy cultivation, small-scale agriculture, and swidden farming. Industrial rubber, cassava, sugarcane, banana, and other plantation trees have been planted by national and foreign leaseholders in lieu of these traditional, smallholder farming systems, often without consulting or seeking the consent of local landholders, and with dire repercussions on their collective food security and food sovereignty.

A similar process of enforced rural relocation has been documented in the arid and drought-prone regions of Northeast Ethiopia, where the establishment of cotton and sugarcane plantations has been accompanied by the development of hydroelectric dams for irrigation purposes, with devastating impacts on the water and land access of Afar and Oromo pastoralists who use the floodplains of the Awash Valley for dry-season grazing. These communities have been displaced to a total of 20,000 and now face severe food insecurity – at times, famine – due to their dislocation from their prior subsistence and grazing grounds (Stevenson & Buffavand 2018). In this and many comparable instances, the forced removal (direct or indirect) of Indigenous Peoples and other rural communities from their homelands has been facilitated by laws that were instituted under colonial rule and whose legacies find expression in current policies that continue to classify Indigenous lands as 'terra nullius' and that either do not recognize Indigenous Peoples' existence or do so on the basis of externally imposed criteria rather than on the principle of self-determination enshrined in international human rights law. p. 11–12

One striking example of this is Indonesia, the world's top palm oil-producing country. Seventy-five years after Indonesian independence, agrarian laws introduced by the Dutch in the mid-eighteenth century, and particularly the legal concept of 'free state domain' (domein verklaring) instituted to promote private investment, continue to leave ambiguous the question of where state claims over land end and community claims begin. This legal opacity radically undermines the tenurial security of rural populations inhabiting or forcibly removed from lands converted to timber, oil palm, and other commodity plantation zones. While Indonesian laws do recognize 'custom' (adat) as a source of juridical authority and 'customary law communities' (masyarakat hukum adat) as a local expression of Indigeneity, these legal instruments remain premised on a system of land rights inherited from the Napoleonic civil codes of continental Europe – one that ultimately weakens pre-existing communal tenures and strengthens the state's prerogative of control by privileging national economic interests over customary and Indigenous rights (Colchester, Chao and Anderson 2024). p. 12

In other contexts, plantation proliferation through Indigenous displacement and rural dispossession is buttressed by prior histories of civil war and guerrilla violence as well as by ongoing realities of para-military occupation. This is the case in Colombia, where intensive deforestation and oil palm expansion are taking place against the backdrop of a perduring armed conflict that has displaced over five million people since 1964. In the region of Bajo Atrato, large-scale plantation developments, purportedly motivated by the expulsion of illegal guerrilla groups, have resulted in the displacement of an estimated 15,000 Afro-Colombian inhabitants, while concomitantly offering a legal basis and justification for counterinsurgency groups to launder capital, strengthen their economic base, and secure their territorial presence in the region (Torres Mora 2020). Across these and many other plantation geographies and histories, the violent removal of Indigenous peoples and rural communities from their native homelands and soils forms a fundamental premise for agro-industrial expansion. Extirpations of people from place to make way for monocrops are intimately tied to the extractivist logic of plantations' own *internal* mechanisms of commodity production, which I turn to next. p. 12–13

2.2 Enslavement and Exploitation

Central to the internal operations of the plantation system is the extraction of labour from workers who were either translocated into foreign agro-industrial topographies under colonial enslavement regimes, forced to migrate and seek plantation employment as a means of economic survival in the absence or loss of alternative livelihoods, or incorporated into agribusiness labour systems established on their own native soils. No period in human history illustrates the first of these processes more vividly than the trans-Atlantic slave trade, which has been recognized by the United Nations as 'the greatest crime against humanity committed in what is now defined as the modern era' (Missouri Sherman-Peter 2022). p. 13

Spanning over 300 years, the trans-Atlantic slave trade of 1501 – 1867 saw some 12 to 20 million African men, women, and children kidnapped, uprooted from their homes, transported on European and American ships as human cargo, and trafficked across the Atlantic Ocean to work under conditions of enslavement in imperial cotton, rubber, jute, tobacco, and sugarcane plantations. Known as the 'Middle Passage', this journey fed and fuelled plantations established by English, French, Dutch, Spanish, and Danish colonists, with chattel slavery instituted through the Slave Code of 1661. Originally developed in Barbados by the English colonial assembly and later adopted as the template

across North American colonies (Rugemer 2013), this Code laid the juridical and moral basis for the enslavement of Africans by defining them as non-human bare flesh and fungible property, readily available for ownership and trade by White purchasers (Spillers 1987; Wynter 2003; King 2016).

p. 13–14 Over 20 per cent of Africans who crossed the Atlantic Ocean aboard colonial ships died of hunger, disease, exhaustion, and abuse before reaching the shores of the United States and the Caribbean. Awaiting those who survived the journey was an egregiously violent regime of labour extraction that was premised on the exploitability of racialized, dehumanized, and commodified Black bodies. This regime was imposed not only on those individuals brought over from the African continent but also on their offspring, with Black women's reproduction subject to enforced breeding regimes that sought to secure plantations' future labour supply (Morgan 2004). Resistance to the abusive conditions of plantation labour in colonial monocrops was routinely met with horrific physical and psychological violence, including public displays of torture, humiliation, and murder (Hartman 1997). In nineteenth-century banana and sugarcane plantations in Central America and Hawai'i, respectively, such resistance was countered through the intentional social division of work across different ethnic groups, which served a divide-and-control function anchored in the conjuring of racialized prejudices, hierarchies, and rivalries between Amerindian, West Indian immigrants, and Central American Hispanics on the one hand (Bourgois 1989) and between East Asians, Norwegians, Puerto Ricans, and Portuguese on the other (Takaki 1983).

p. 14 In a devious instrumentalization of perceived human-vegetal affinities, labour extraction on colonial plantations often operated on the assumption that enslaved or exploited people were best suited to the conditions of plantation work and the proclivities of the particular plants cultivated. In eighteenth-century Kentucky, for example, hemp was often described as a 'nigger crop', its biotic recalcitrance, marshy growth environments, and colossal labour requirements said to be ideal work for 'negros', who were in any case as 'dirty' as the process required to manage and process the plant itself (Hopkins 1951: 24). These beliefs continue to undergird contemporary plantation labour regimes, wherein racial traits and social-evolutionary ideologies are deployed to justify or naturalize the exploitation of some populations over others. In the Paraguayan soy sector, for instance the unwieldy plant of cotton is frequently described as a crop best cultivated by brown peasants or 'campesinos', and the modern plant of soy by White people (Hetherington 2019). In the Ecuadorian flower sector, meanwhile, participation in plantation labour is often framed by agribusiness operators as a means of uplifting and civilizing Indigenous peoples from their primitive ways of life, in a form of racial emancipation and liberation (Krupa 2022). This dynamic echoes in uncanny ways the discourses of agronomists and government officials in Malaysia at the turn of the twentieth century, wherein the scientific care of oil palm plants and plantations was thought to presuppose care of the 'native', providing a necessary corrective to what were percieved as 'careless' local modes of cultivation and production (Rudge 2022).

p. 14–15 In other contexts, the relocation of people from their native homelands to industrial plantations is less a process imposed by agribusiness operators or the state, but rather a means of economic survival sought out by precarious communities themselves and often tied to the promise of wealth and opportunity. A case in point is the Malaysian oil palm industry (formerly the world's largest palm oil-producing country) that relies on an estimated one million migrant workers employed on estates and mills in Peninsular Malaysia, Sarawak, and Sabah. Most of these workers hail from Indonesia, notably the less-developed islands of Sumatra, Sulawesi, Lombok, Sumbawa, and Flores. Many eke out a precarious existence dependent on insecure, casualized, and flexibilized work arrangements including sub-contracted labour and low/piece-rate wages. National laws prohibit migrant workers from settling permanently in Malaysia, acquiring Malaysian citizenship, influencing Malaysian culture, marrying Malaysians, or sending their children to local schools. Workers' captivity to plantation employers' biopolitical control is further exacerbated by the confiscation of their passports, debt bondage, the selling and reselling of migrant workers by brokers, and the risks of deportation, physical punishment, and imprisonment (Pye et al. 2012). These dynamics position agribusiness plantations as just one of many manifestations of racialized regimes of extraction globally ‑from clothing and technology manufacturing sweatshops to the industrial livestock and dairy industry, oil, gas, mineral, and coal mining, and many other sectors sustained through relations of exploitative labour ‑that might not constitute literal monocrops, but that nonetheless trace their origins back to plantation logics and that continue in fields and factories around the world (Chao, Wolford, Ofstehage et al. 2024, 13).

p. 15 The extractive logic of plantation labour has long been paired with its attritive effects on the physical and reproductive health of agribusiness workers ‑a reality that comes to light vividly in the context of prevalent pesticide use within the monocrop sector. Particularly prominent among these non-human chemical actors is paraquat, a carcinogenic herbicide that has been linked to kidney and liver failure as well as Parkinson's disease. Banned in over fifty countries, paraquat continues to be used in plantations across the Global North and South, notably in palm oil operations in Southeast Asia. Another prevalent chemical is dibromo-3-chloropropane, or DBCP, the active ingredient in the nematicide Nemagon, which has caused thousands of banana plantation workers in Costa Rica and Honduras to suffer from severe and long-term health conditions include sterility and cancer (Soluri 2005). Elsewhere, including in Martinique and Guadeloupe, it is the pesticide chlordécone ('kepone' in English) that poisons the bodies of banana plantation workers, alongside the islands' soils and sources of water. Originally produced in the United States to combat the effects of the parasitic

banana borer beetle, this hormone-altering substance is a major driver of male infertility and prostate cancer among workers whose exposure to the toxin is aggravated by the lack of protective equipment or sanitary measures (Agard-Jones 2013).

In California's strawberry plantations, meanwhile, pathogenic control through regular fumigation and consequent pesticide drift is a primary cause of breathing difficulties, nausea, burning, and headaches among Latino migrant workers, who are ineligible for income support until they are completely disabled and who are reticent to call out pesticides' adverse health effects for fear of being fired or deported (Guthman 2019). In these and other instances, the promotion of ill-regulated chemical use in plantation landscapes points to a privileging of cash crops' health and productivity over the physical and intergenerational wellbeing of plantation labourers, whose extractive potential is paired with their inherent disposability and replaceability in the generation of capital and surplus –all within a broader system that relies on and maintains global as well as local wealth and racial inequalities. But humans are not the only ones subject to such extractive regimes. So too are plants themselves, as objects of biotechnological manipulation and forms of lively capital.

p. 15–16

2.3 The Labour of Plantation Crops and Ecologies

To think about plants as objects of extraction is to draw attention to the metabolic labours that vegetal beings perform in generating the products of plantation economies –whether it be their fruit, sap, trunks, nuts, or seeds. Sustaining these non-human metabolic labours requires an understanding of plants' processes of growth and senescence, their photosynthetic affordances, their vegetative or sexual reproduction, as well as the soil, water, nutrient, and climate requirements needed to support intensive and large-scale cultivation. While the metabolic labours of plants might be anchored in empirically grounded, scientific epistemologies including botany, agronomy, and biology, they have also been widely harnessed to further racial configurations of plantation labour and colonial-capitalist technologies of land expansion and yield intensification. Attending to more-than-human forms of work thus matters in uncovering what cash crops demand of humans in terms of management, nurture, and care. But it also sheds light on how humans have instrumentalized plants' biotic and ecosystemic affordances to legitimize their manipulation or transformation in the name of productivity and profitability, and to buttress regimes of human exploitation, social inequality, and racial violence.

p. 16

Take the cotton plantations of the nineteenth-century Southern United States. The high nutrient and soil demands of large-scale cotton plantations, identified in scientific studies of the time, together with the crop's notorious vulnerability to pests and parasites, were at first remedied through the acquisition of ever-greater swaths of land and subsequently through the importation of guano and the introduction of synthetic pesticides and fertilizers. The recognition that intensive plantation agriculture depleted soil quality later served to justify the deployment of enslaved workers not only to work in plantations but also to mine marl (a sedimentary rock containing lime) without payment or compensation. In this 'chemicalization of slavery', the productive value of enslaved people was equated to the fertility-boosting qualities of inanimate matter, including soils, minerals, and faeces (Williams & Porter 2022: 505).

p. 16–17

In the colonial Caribbean, it was the perishability of sugar –notably its high susceptibility to degradation through fermentation –that legitimized alterations to processing technologies and divisions of labour. This included the association of racialized bodies with specific tasks in the sugar production chain and the pursuit of biochemical purity through the exploitation of Black people, who were paradoxically tasked with whitening sugar by purging it of dark impurities through their enforced and often dangerous labour in mills and refineries (Rood 2017). In mid twentieth-century Liberia, meanwhile, the extraction of rubber –or 'white gold', as it was commonly referred to –relied on Black tappers' intimate understanding of how deep and wide to cut the trunk in order to extract the sap without damaging the tree's regenerative abilities (see Figure 2). Failure to achieve this would lead to loss of employment and corporal punishment for the labourers tasked with the tapping (Mitman 2021).

p. 17

Plants' biotic affordances also come to determine human labour patterns in contemporary plantation systems. In tea monocultures in India, for instance, workers engaged in continuous leaf plucking throughout the year are necessary to meet the continuous sprouting of tea leaves in rhythms of 'flush', a pause which causes sprouts to mature and therefore spoils the quality of the final product (Barua 2023). In the Southeast Asian palm oil sector in contrast, ad hoc, casualized, or seasonal work arrangements that involve low unit labour requirements and create economic precarity are driven by the slow production cycle of the oil palm plant itself, whose fruit yield does not peak until the ninth year, requires harvesting every two weeks, and begins to decline around the fifteen-year mark. For oil palm smallholders, meanwhile, the imperative to process fresh fruit bunches within forty-eight hours of picking to avoid oil acidification and spoilage often positions them in monopsonistic relations with the mills that they depend upon to process their harvest, but that are owned and operated by private corporations (Cramb & McCarthy 2016).

p. 17–18



Figure 2 Close-up of a rubber tree trunk with a diagonal tapping cut. Image credit: Lucia Barreiros Silva.

p. 18 In other cases, optimizing the extraction of profit from cash crops has prompted human-driven modifications of plantation ecologies through the introduction of so-called 'beneficial species', whose activities play a key role in enhancing plant health and productivity within biological pest and parasite management programmes. For example, the damaging effects of rat infestations in Indonesian oil palm plantations are now increasingly controlled through the introduction of predator species, including leopard cats, Asian palm civets, snakes, and diurnal birds such as the yellow-bellied prinia, lesser coucal, ashy tailorbird, great tit, Oriental magpie-robin, and barn owls, which are bred in rearing centres and then released into the plantations. The deadly effects of the parasitic rhinoceros beetle are kept at bay through the actions of yellow assassin bugs, tachinid flies, and chalcidoid wasps that prey on the beetle's eggs, and the introduction of entomopathogenic microorganisms, including various fungi, bacteria, and nonoccluded RNA viruses, that attack the epidermal cells of the beetle's larvae (Chao 2021a). Comparable enlistings of mutualistic non-human organisms are at play within coconut plantations in the Philippines, where native kiwot bees are intentionally bred to stimulate the pollination of coconut trees, with a reported 80 per cent enhanced yield (Conde 2020). These plantations are also home to cattle herds that are locally known as 'sweepers' or 'brushers' and that, under the so-called 'cattle under coconuts' system, provide key ecological services by grazing on weeds, grasses, and undergrowth, thus preventing nutrient and moisture competition with coconut palms and naturally fertilizing the soil with their manure (Plucknett 1979).

p. 18–19 An arguably more controversial means of enhancing the extractive potential of cash crops than the ecological engineering described above takes the form of genetic modification (GM), a practice that took root in the late twentieth century following the discovery of DNA structure and subsequent scientific experimentation in gene cutting and recombination. Following the 'Green Revolution' and the inception of large-scale agriculture in soybean, maize, cotton, canola, and other plantation commodities across the Global South, transgenic crops were developed for the purposes of achieving stronger herbicide tolerance and pest resistance, and later came to be widely promoted in agronomic and policy discourse as key to sustaining the food security of developing nation-states (Conway 2012). Their proliferation, however, has and continues to be contested on multiple grounds.These include GM crops' potentially harmful impacts on consumer and environmental health, their detrimental effects on indigenous plant and agricultural intellectual property rights and patenting opportunities, their adverse consequences for small-scale farmers' economic empowerment and market access, and their implications for plants themselves as lifeforms subject to ethically questionable and primarily corporate-led agri-bio-technological manipulation and homogenization.

p. 19 The complex entanglements of science and democracy at the heart of the GM tide come to light in India, where insect-resistant Bt cotton introduced in 2002 by the American agrochemical and agricultural biotechnology company Monsanto in partnership with the local firm Mahyco now occupies over 90 per cent of local farms, with negative effects not only on small and marginal farmers' economic independence from global and national seed corporations companies but also on cotton's own resilience resulting from the substitution of sturdier, older varieties with more fragile, longer-staple hybrids (Aga 2022). In Argentina, where genetically modified soybeans cover half of the country's arable land and constitute a third of total exports, GM crops are celebrated in public discourse as heralds of modernization and economic growth within the so-called 'soy boom'. This discourse, however, sits alongside stark realities of social and environmental injustice. Power, land, and profit are increasingly concentrated in the hands of a small agribusiness elite, and traditional crops are systematically replaced with plantation commodities, with dire consequences for local food security, forest ecosystems, rural livelihoods, and the health of communities exposed to the agrochemicals required to

combat GM soy's growing herbicide resistance (Leguizamón 2020).

In other instances, the implications of GM crops for human health, together with their degree of cultivability in different ecological milieus, have long been under question. This is evidenced in the Philippines through the case of the much-promoted Golden Rice, a beta carotene-enhanced strain that promised to curb Vitamin A deficiency-related diseases and save millions of lives in the Third World but that remained at the trialling and pre-regulatory stage for almost a quarter of a century until 2022, when it first entered commercial production. Following this development, civil society organizations including Greenpeace and local farmer collectives, voiced concerns to the Supreme Court over the threat of biodiversity contamination and economic marginalization posed by the genetically engineered and patented crop to the estimated 7,000 unique traditional rice varieties grown, consumed, and sold by independent smallholders across the country's biodiverse rice landscape. Given the existing and anticipated impacts of climate change, these actors also highlighted the serious risks posed by the reliance on a single-crop system for communities' socio-economic resilience (Stone & Glover 2016; Fabro 2023). The appeal led to a court decision in May 2024 that halted the cultivation of Golden Rice in the Philippines. While celebrated by its proponents, the decision subsequently prompted widespread condemnation over the tens of thousands of children's deaths that would potentially result from the blocking of what some segments of the scientific community hailed as the 'lifesaver' crop (McKie 2024) – even as on the long-term, monocrop plantations have proven a severe threat not only to local ecosystems, but also to the food security, dietary health, and nutritional wellbeing of local communities (Chao 2025). p. 19–20

2.4 Resistance and Failure

The Golden Rice controversy described above offers a segue into another important facet of plantation extractivism – namely, the ways in which extractivism is resisted at the grassroots and beyond, and the ways in which it can fail despite the seemingly relentless and ubiquitous force of technocapitalism and attendant state-corporate agendas. Indeed, the naturalization (in literal and figurative terms) of plantation logic, while indisputably pervasive and powerful in its reach and effects, has never gone entirely uncontested by those subject to its violence, nor have its extractive machinations always run as smoothly as might appear on the surface (Wolford 2021; Chao, Krupa & Li 2023). Human collectives have forged alliances in the face of plantation occupation, sometimes spanning countries and continents in terms of their membership, their activities encompassing concerted protest, multi-pronged advocacy, and strategic negotiation, as well as everyday, passive forms of resistance, including foot-dragging, theft, sabotage, non-compliance, and carefully cultivated inefficiency (Stoler 1985; Peluso 1992; Lewis 1996). Non-human entities, too, have in perhaps less intentional ways undermined the plantation project through their biotic affordances, including parasites that compromise the yields and productivity of monoculture cash crops rendered vulnerable to pathogenic attacks as a result of their genetic homogeneity. Perhaps most striking among these forms of resistance are those that involve the emergence of interspecies solidarities in the face of plantation extraction. In these emergences, the agendas of exploited humans and non-humans have aligned and become the unexpected grounds for new configurations of plantation subjecthood, victimhood, justice, and repair that are anchored not in rigid speciesist divides but rather in a counter-hegemonic blurring of animality, vegetality, and humanity. p. 20

In some instances, it was the ecologies within and upon which plantations were established that pushed against colonial visions of order, discipline, and homogeneity. In the American plantation zone, for instance, the prevalence of dense tropical vegetation and impenetrable swamplands, bayous, rivers, and wetlands often proved quasi-impossible for imperial powers to successfully map or control. Heat, diseases, vines, snakes, and insects joined forces with ancestors, spirits, and humans as a collective, animate force that repurposed the putative animality of exploited, racialized bodies into a source of resistance and revolution, and that became the birthing ground for enduring botanico-religious cosmologies including Obeah, Vodou, and Santería (Allewaert 2013). These unruly, more-than-human, and more-than-natural ecologies significantly impeded the productivity of imperial ventures at the same time as they became precious shelters for diasporic Africans fleeing capture and enslavement, generating a fleeting Black commons and precious multispecies refuge from the topographies of plantation violence (Hosbey & Roane 2021). p. 20–21

Other interspecies solidarities arose within the confines of the plantation itself. In Jamaican sugarcane fields, for example, sorghum shoots were celebrated in the songs of enslaved Africans and became active, non-human participants in scenes of artistic and ecological commoning alongside tea, water, yams, and provision grounds (Dillon 2019). In the colonial indigo plantations of South Carolina and the Low Country, dark dyes seeped deep into the hands of enslaved women labourers, merging with fungible Black bodies in the making of porous part-human, part-vegetal flesh and generating new multispecies companionships, forged amidst and against the dehumanizing violence of the plantation system (King 2019: 111–140). On plantations across the Caribbean and the American South, food plots sown with root crops, plantains, bananas, and medicinal herbs, and cultivated by the enslaved, became unexpected biocultural havens that nurtured oppositional modes of Black life. The cultivation and celebration of these plots were anchored in understandings of the Earth and of the land as a goddess, bound in relations of reciprocal nourishment with humans and bringing humans and non-humans together in socio-ecological and cosmo-spiritual assemblages of kinship, repair, p. 21–22

and care (Wynter 1971; Trouillot 2002; Davis et al. 2019). In other instances, non-humans became precious allies in processes of flight and fugitivity, or what has come to be known as marronage. These included rice seeds, black-eyed peas, cassava cuttings, and many other vegetal organisms that enslaved women in Suriname, French Guiana, Brazil, and the broader Caribbean carefully concealed in their braided hair before Trans-Atlantic crossing (Carney 2004; Keeve 2020; van Andel, Maat & Pinas 2023). Women then planted these seeds to feed themselves and their children during their escape from colonial plantations, their actions playing a vital role in supporting the food security and social continuity of newly formed escaped slave communities, or Maroons, and also transforming the broader ecological landscapes within which these communities gained ground and grew.

p. 22 More-than-human alliances continue to animate plantation landscapes of the present, including in ways that blur the line between the material and the metaphorical. In the Argentinian soy sector, for example, activists invoke resonances, similarities, and contiguities between the resistance of stubborn weeds like amaranth to the herbicide glyphosate and their own collective struggle to oppose the expansion of GM crops. Here, the resilience of amaranth lends momentum to human protests that come to mirror the plant's own refusal to comply with the agendas of genetically engineered cash crop projects (Beilin & Suryanarayanan 2017). A similar plant-human indexicality is at play in the Papuan oil palm industry, where parasites, including rats, porcupines, coconut caterpillars, the Ganoderma fungus, and the Rhinoceros beetle, are often celebrated by Indigenous anti-plantation activists as more-than-human arbiters or agents of retributive justice for the damages inflicted by the relentlessly expanding agribusiness sector, their slow, often invisible, yet highly effective actions in turn inspiring human resistance to oil palm monocrops and their colonial-capitalist undergirdings (Chao 2021b). In the Indian tea sector, meanwhile, workers' resistance to the surplus and unrecognized labour and time required of them to achieve the demands of organic cultivation both affects and is affected by non-human agency. Intentionally inconsistent tea-plucking on the part of workers, for example, results in an overgrowth of stubborn yet edible fiddlehead ferns. This generates unruly ecosystems that defy the control and discipline of the plantation regime, at the same time as it creates precious foraging grounds that support the food security and nutritional diversity of plantation labourers (Kumpf 2021).

In other cases, non-human partners involved or invoked by people affected by monocultural extractivism transcend the realm of the biotic and material, pointing to the more-than-secular dimensions of plantation lifeworlds. In Eastern Borneo, for example, spirit communication and customary healing rituals have played a major role in sustaining a sense of self and sociality among Indigenous Luangan communities, whose environments have been radically transformed by oil palm but who find, in encounters with spirits, a powerful means to alleviate everyday experiences of unpredictability and alienation (Hermanns 2020). In a comparable vein, alliances forged with outsiders by Indigenous Makushi communities in Guyana to oppose rubber expansion operate in conjunction to long-standing relationships with shamans, whose ability to communicate with and harness the powers of spirits lends a powerful, cosmological dimension to Makushi struggles for justice in the face of ongoing land dispossession and occupation (Whitaker 2020).

p. 23 The potency of more-than-human resistance to, or non-compliance with, the extractive visions of plantation logic manifests perhaps most strikingly in instances where monocrop projects have collapsed entirely or been abandoned due to unsurmountable environmental and ecosystemic factors. In Brazil, for instance, efforts to defeat the South American Leaf Blight ‑a type of fungus that has proved remarkably resistant to chemical treatment, grafting, and plant breeding ‑have been consistently ineffective, in turn shattering the country's visions of establishing itself as a leading industrial rubber producer (Dean 1987). Relentless attacks by mites, lace bugs, caterpillars, and various other endemic tropical critters were also partly responsible for the dramatic collapse of 'Fordlandia', a grandiose, if entirely unrealistic, project initiated by American business magnate Henry Ford in the early twentieth century and aiming to bring plantation-scale rubber production to the Amazon jungle, alongside the construction of midwestern-style towns, hospitals, railroads, power plants, golf courses, and tennis courts. After almost twenty years of thwarted efforts to control an environment completely unfit for monocrop rubber cultivation, the entire operation was sold by Ford to Brazilian buyers for a tenth of what it had cost to establish (Grandin 2010).

The dramatic consequences of neglecting ecological factors have more recently been exemplified by the 1997‑1998 Indonesian Mega Rice Project in Central Borneo. Initiated by the government and aimed at transforming one million hectares of peat swamp forest into rice paddy fields, the project was, from the outset, marred by the nutrient-poor nature of local peat soils, the unsuitability of Borneo's ecosystems for wet rice cultivation, and the struggles experienced by the imported labour force to manage an unfamiliar environment. The disturbance and drainage of tropical peat soils, compounded by a year of extreme drought during the El Niño event, resulted in peat fires spreading uncontrollably under and overground and generating in the space of a few months the same amount of carbon dioxide as an estimated 13‑40 per cent of global annual carbon emissions from fossil fuels (Goldstein 2016). Abandoned within two years, the Mega Rice Project has been recognized as one of the most devastating environmental disasters in Indonesia's history, at the same time as it produced almost no rice.

2.5 Conclusions

The framing of the 'Plantationocene' outlined in the Introduction to this Element offers a useful analytic for thinking about extraction in multispecies terms. Rather than centring (and singularizing) the 'human' as the driver of ecological transformation (as in the 'Anthropocene') or relying on an abstracted notion of 'capital' as the root cause of environmental decline (as in the 'Capitalocene'), the Plantationocene invites us to take seriously the complex entanglements of particular plants, particular plantations, and particular people within extractive landscapes, together with the congeries of species, symbiotes, and spirits that also partake in the making and unmaking of monocrop systems and societies. As I have demonstrated in this section, multispecies extraction has long been a defining feature in the rise (and, at times, fall) of plantation lifeworlds –from the forced removal and displacement of Indigenous Peoples and local communities from their ancestral and more-than-human territories, to the systematic extraction of labour from racialized bodies, themselves often uprooted from native soils, to the increasingly technologically mediated and engineered extraction of cash crops' own biotic vitality as seeds of life and sources of profit. This multi-faceted, extractivist logic is premised on colonial-capitalist ideologies that assume the malleability of environments and the compliance of plantation subjects, as well as a rigid division of nature from culture.

p. 23–24

And yet, as the final section of this section has shown, these dangerous fictions are belied by realities of resistance to plantation violence, many of which are grounded in interspecies indexicalities that muddy the boundaries between the human and the non-human as subjects of harm and agents of defiance. Attempts to simplify, control, and conjure plantations as capitalist natures have often failed due to ecological factors and biotic actors whose disruptive affordances are all too often underestimated in the design and implementation of anthropogenic landscape transformations (Scott 1998; Tsing 2015; Tsing, Mathews & Bubandt 2019). These factors and actors may not be considered intentional –and yet, they have often been harnessed and signified as such by human collectives within counter-plantation imaginaries, struggles, and solidarities. Retrieving sovereignty in the plantation, as such, has often entailed for communities the world over, a form of rallying-with the non-human in its myriad forms, anchored in a recognition of shared vulnerabilities and threatened futures across –and sometimes beyond –species lines.

p. 24

3 Extinction

p. 24–25 Extinction constitutes our second analytical entry point into plantation lifeworlds and afterlives. While intrinsically connected to extraction in terms of its dynamics and effects, extinction draws attention to the forms of *destruction, extermination, loss, and ruination* that have historically accompanied monocrop proliferation and whose legacies remain embedded in the landscapes of the present ‑even where these landscapes may no longer be classified or exploited as agro-industrial zones. In this section, I examine how plantation-driven species and ecosystemic extinctions sit within, and further, a broader process of ecological decline and climate crisis, wherein industrial and human activities are undermining conditions of life at a planetary scale (Chao & Kirksey 2022). I consider how environmental destruction erodes the intimate and ancestral biocultural relations and multispecies histories binding peoples, plants, and places. In doing so, I follow the lead of environmental humanities scholars in approaching extinction not as a one-off 'event', marked by the passing of the last individual of a species, but rather as a loss that sits within broader patterns of unravelling and that entails the erasure of ways of life shared across species lines ‑including with humans (van Dooren 2014; Rose & van Dooren 2017). In closing, I consider the promise and perils of corporate sustainability and conservation initiatives that seek to greenify the plantation sector but run counter to the ways of knowing, being, and relating that communities entertain with non-human lifeforms and landscapes, thus perpetuating, in a different guise, the colonial logic of the plantation itself.

3.1 Biodiversity Loss and the Climate Crisis

p. 25 As production systems that often entail major land-use changes, industrial plantations constitute primary drivers of deforestation, biodiversity loss, and the erosion of vital ecosystem functions including carbon storage, soil regeneration, and air and water quality. Their proliferation drives almost 50 per cent of global forest loss, with carbon released into the atmosphere from plantation-driven deforestation accounting for around 10 per cent of global warming emissions. These impacts are symptomatic of, and directly contribute to, the broader phenomenon of climate change, which is reconfiguring the terrains and topographies of multispecies existence across an increasingly vulnerable and volatile planet.

p. 25–26 The oil palm industry is a case in point. Oil palm, or *Elaeis guineensis* Jacquin in Linnean taxonomy, thrives best in humid lowland environments, and particularly in riverine forests and freshwater swamps. Shade-intolerant and hydrophilic, the plant requires high levels of soil moisture, temperatures ranging from 24 to 32 degrees Celsius, and at least 1,780 to 2,280 millimetres of rain per year. These factors combined make the planet's tropical belt the ideal environment for oil palm cultivation. However, it is also in these same tropical zones that the highest levels of biodiversity today are to be found, and where the most rapid rates of species extinction are underway (Qaim et al. 2020; Chao 2022d). Today, palm oil is one of just four commodities driving the majority of tropical deforestation (up to half, according to some reports) and oil palm plantations represent the second largest driver of global warming after beef production (Union of Concerned Scientists 2016). Between 2000 and 2010 alone, oil palm monocrops were responsible for 2‑9 per cent of worldwide emissions from tropical land use (Carlson & Curran 2013). Most of these emission stem from Indonesia and Malaysia, who together produce over 80 per cent of the world's palm oil supplies.

p. 26 The establishment of plantations on carbon-rich peatlands in Southeast Asia plays a major role in the story of oil palm-driven multispecies extinctions and ecosystemic destruction. Peatlands may release up to twenty-eight times as much carbon as the forests above them when drained for agricultural purposes, requiring more than 600 years for the carbon balance to be re-established (Danielsen et al. 2009). Easily ignited following drainage and difficult to extinguish, predict, and control, peat fires can last for weeks or even months, smouldering subterraneously over vast areas and sometimes re-emerging miles away from their initial source (Page & Hooijer 2016). The flammability of peatlands is exacerbated by concentrated cash crop planting, which over time reduces freshwater and soil quality, undermines critical ecosystem services once sustained by forests such as regulation of the hydrological cycle and soil protection, and overall enhances the risk of abiotic effects, including intensified wind, desiccation, and fire risk. Gradually, soil erosion and nutrient and mineral depletion undermine the very grounds of the plantation, together with the other-than-human communities of life that inhabit and depend upon them.

p. 26–27 The simplified ecology of monocrop oil palm plantations proves a largely inhospitable environment for those organisms who manage to survive the felling and flames. This ecology is characterized by uniform tree species and tree age structure, low canopies, sparse understory vegetation, low-stability microclimates, the almost-complete absence of leaf litter, a twenty-five to thirty-year clearing cycle, and run-off from chemical fertilizers and pesticides (Obidzinski et al. 2012; Petrenko et al. 2016). Habitat fragmentation, the shift to simpler community compositions, the absence of ecological corridors, and the impervious barriers posed by plantations to animal migration undermine both species richness and abundance, allowing only about 15 per cent of native vertebrate animal species to survive the transition from primary forest to plantation, and fewer than half of vertebrate communities (Fitzherbert et al. 2008; Savilaakso et al. 2014). As a result, palm oil production is adversely affecting at least 193 threatened species globally, according to the

International Union for the Conservation of Nature Red List of Threatened Species. These species encompass 54 per cent of all threatened mammals and 64 per cent of all threatened birds globally (Meijaard et al. 2018). Among the most vulnerable to forest-monocrop landscape transitions and associated heightened poaching activities and human-wildlife conflicts are the critically endangered Bornean and Sumatran orangutans, the Sumatran tiger, the Sumatran rhinoceros, and the Sumatran and Bornean pygmy elephants, alongside forest specialist species such as the muscicapine flycatcher, the arboreal fruit-eating gibbon, and the peat swamp frog.

The soy industry is another key driver of biodiversity loss, its expansion contributing to 8.2 million hectares of forest replacement between 2001 and 2015 and representing over 10 per cent of tropical deforestation tied to agricultural commodities during that period (World Resources Institute 2024). The proliferation of this crop is particularly marked in Latin America, notably in Brazil, Argentina, Paraguay, Bolivia, and Uruguay, where over half of the world's soybean production takes place, and where the perceived availability of cheaper tropical lands is legitimating the expansion of soy as a source of beans, oil, and meal for human and livestock consumption, and as a source of renewable energy (Song et al. 2021). p. 27

In Brazil, soy plantations are concentrated in ecologically sensitive regions such as the Amazon, where mass-cultivated areas increased fourfold from 1.14 million hectares in 2006 to 4.48 million hectares in 2016, and the Cerrado, which is home to some of the most biodiverse savanna landscapes in the world (Sauer 2018; Schneider et al. 2021). Unlike the Amazon, the Cerrado lies outside the zones protected by the 2006 Soy Moratorium, a voluntary pact intended to reduce deforestation caused directly by soy (Brandão, Cardoso & Garrett 2025). The region has suffered a 50 per cent loss in native vegetation due to monocrop developments in the last two decades, leading to the critical endangerment of thousands of endemic plants and animals, including the maned wolf, giant anteater, blue-eyed ground dove, and white-nosed saki. In tandem, slash-and-burn methods used to clear land for soy contribute to degraded air quality and greenhouse gas emissions, at the same time as they render fragmented forests increasingly vulnerable to fire. The leaching of agrochemicals into rivers, waterways, and aquifers, combined with the reduction in evapotranspiration resulting from forest felling, is undermining aquatic biodiversity, degrading native soils, and causing unprecedented fluctuations in the moisture and rainfall cycles essential to multispecies flourishing across the Cerrado, the Amazon, and well beyond (Rekow 2019).

Often, oil palm and soy expansion intensify, rather than create, the threats of extinction faced by tropical flora and fauna because they unfold within a broader mosaic of extractive industries, past and present. These industries include other agribusiness ventures, but also mining, logging, and pulp and paper operations, and infrastructural developments such as roads, railways, artificial waterways, hydroelectric dams, and ports, which together compound the vulnerability of ecosystems that endangered species depend on to survive and thrive. In the case of soy in the Brazilian Amazon, for example, and despite policies instituted under the aforementioned Soy Moratorium, agribusiness developments continue to drive indirect deforestation by establishing themselves on former cattle ranches and grazing areas that are then forced to move further inland. Forest fragments left standing between developed zones tend to be small, sparse, and susceptible to adverse and cumulative edge effects. The lack of biotic corridors connecting these fragile ecological patches often contributes to undermining their ecosystemic viability, including by hindering species migration and dispersal, and pollination through biotic vectors. p. 27–28

The extinctions and endangerments generated by the 'modular simplifications' of biodiverse landscapes turned agribusiness monocrops often work hand in hand with 'feral proliferations' that are directly linked to the genetic homogenization of plantation ecosystems (Tsing, Mathews & Bubandt 2019). For while on the short term, monocultures might be advantageous in terms of control and productivity, on the long-term, they are more vulnerable to the stresses of disease than polycultures, their uniformity creating a favourable habitat for pests specialised in predating on the particular species cultivated. p. 28

In Mexico, for instance, the intensification and technologization of coffee cultivation, combined with mass deforestation and a resulting increase in wind-borne spore travel, triggered a massive outbreak in 2012–2013 of coffee leaf rust, a disease caused by the fungus *Hemileia vastatrix*, which threatened to devastate coffee production across Central America, the Caribbean, and Colombia (see Figure 3). This epidemic has heightened the likelihood of future and more frequent outbreaks, especially given that rust-resistant varieties are vulnerable to other fungal diseases, such as ojo de gallo, and that many are already showing signs of rust infection (Perfecto, Jiménez-Soto & Vandermeer 2019). The susceptibility of monocrops to parasitic activity also comes to the fore in Honduras, where the shift in banana's role from agroecological shade plant to industrial cash crop, dominated by a single variety (the Gros Michel), led to the rampant spread of Panama Disease (mál de Panamá, mata muerta, or *Fusarium oxysporum f. cubense*), which became one of the most economically significant epidemics of the twentieth century (Soluri 2000, 2005). Dense plantings and the homogeneity of the dominant plant, combined with the development of plantation-supporting infrastructures including irrigation canals, drainage ditches, and railroads, all conspired to facilitate the pathogen's spread across the country and beyond. In Liberia, meanwhile, land clearing and the destruction of native flora and fauna to make way for rubber plantations in the early twentieth century allowed opportunistic insects to thrive, with hugely harmful impacts not only on plantation operations but also on workers' health, wellbeing, and safety (Mitman 2021). One such parasite p. 28–29

was the biting black fly (*Simulium* spp.), that found an ideal home in plantation waterway infrastructures, wind breaks, and shade and that preyed on the blood of rubber tappers. Another was the worm *Onchocerca volvulus*, that found a vector both in black flies and in the bodies of the rubber sector's imported labour force, and that became one of the leading causes of blindness in sub-Saharan Africa and other regions of the tropics.



Figure 3 Coffee plant branches laden with clusters of ripening cherries. Image credit: Quang Nguyen Vinh.

p. 29–30

When it comes to cash crops like soy and oil palm, alongside jatropha, sugarcane, sugarbeet, sunflower, rapeseed, and corn, the great irony underlying multispecies extinctions, ecological destructions, and feral proliferations is that plantation expansion (together with associated deforestation and habitat degradation) is often driven by imperatives and discourses of sustainability, which are themselves in turn propelled by the urgency of transitioning from fossil fuel economies to renewable energy sources (Dauvergne & Neville 2010). Debates rage on within policy and scientific circles as to whether biofuel production calls for plantation expansion over greater areas of land or production intensification over limited territories (Byerlee, Stevenson & Villoria 2014). Proponents of the former argue that expansion helps diminish plantations' ecological and ecosystemic impacts by distributing them out across larger geographies. Proponents of the latter, on the other hand, posit that intensification helps concentrate these impacts in more localized and controlled zones. Similarly, emerging progressive policies such as the Soy Moratorium have, on the one hand, been praised for supporting the preservation of exceptionally biodiverse ecosystems, and, on the other hand, criticized for legitimizing the transformation of biomes lying *outside* regulations' protective ambit into 'sacrifice zones' that are consequently rendered prone to heightened and ill-regulated exploitation (Oliveira & Hecht 2016).

p. 30

The actual implementation and on-the-ground effectiveness of such policies and initiatives, meanwhile, can all too quickly be undermined or overturned as government priorities and global markets shift. In Brazil, a 2019 decree passed by then-President of Brazil, Jair Bolsonaro, lifted a ten-year ban on sugarcane cultivation in the Amazon rainforest and the Pantanal (the largest floodplain in the world and a UNESCO World Natural Heritage Site), thus legitimating agribusiness developments in an already highly vulnerable ecosystem (Lima et al. 2020). In Indonesia, a statement by newly elected President Prabowo Subianto in early 2025 called for the massive expansion of oil palm plantations to feed domestic food and biofuel demands, signalling the potential reversal of a moratorium initiated in 2011 on the issuance of new oil palm plantation permits and licences (Suhardi 2025). In these and many other instances, we see that efforts on paper and in practice to protect biodiversity within the plantation nexus are never set in stone, nor guaranteed to survive the volatile flux of shifting (geo)political climates, currents, and concerns.

3.2 Multispecies Lifeworlds and Histories

The ecological destructions and disfigurations described in the previous section are inflected by local experiences, knowledges, and imaginaries that speak in different ways to how plantation developments reconfigure multispecies relations and lifeworlds, alongside biocultural histories and landscapes. What emerges from attending to these local onto-epistemologies is a bottom-up, nuanced understanding and appreciation of the deepentanglements of humanand non-human fates and futures in the face of monocrop expansion –entanglements that challenge assumptions of a neat divide between nature and culture in the context of anthropogenic formations like monocrop plantations.

The terrains of Central Italy offer a compelling example of such entanglements and their layered, temporal dimensions. Here, local inhabitants read in the textures and terrains of the landscape multiple intertwined histories of changing multispecies interactions and inheritances –from the destruction of chestnut orchards that their families cultivated and managed for hundreds of years by pathogens introduced through international trade in the mid-nineteenth to mid-twentieth centuries, to the expansion of state-driven pine plantations in the context of rapid capitalist expansion in the early nineteenth century, which transformed rural peasants and shepherds into industrial workers (Mathews 2018). As they walk, notice, draw, and describe the landscape, local farmers and foresters identify in the morphology of trees the traces of their prior relationships with humans, diseases, fires, and soils, and connect these in their discourses to history, politics, state-making, and broader processes of environmental change. They mourn the enforced abandonment of traditional crop varieties, together with the loss of related practices, including litter raking and controlled burning, that accompanied the rise of conifer concessions. They express deep concern over the dramatic increase of fires resulting from the expansion of the fire-prone species *Pino marittimo*. At the same time, they draw on multispecies histories embedded in the landscape to envision and enact other kinds of futures for themselves and their non-human companions –from efforts to preserve low-elevation chestnut varieties that are more resilient to climate change, to initiatives to support the livelihoods of formerly displaced and ongoingly precarious peasant firewood cutters. p. 30–31

Intertwined histories of more-than-human violence and vulnerability are also at play in the Argentinian Gran Chaco, where large-scale soy fields are being developed today upon landscapes formerly occupied by sugarcane monocrops (Gordillo 2004, 2014). Indigenous communities including the Toba People connect these consecutive histories of plantation proliferation to a range of material, ecological, and affective ruptures that have over time severed them from their intimate and ancestral relations to the landscape, and that materialize not only in soy fields but also in the rubble and debris of colonial infrastructures that were tethered to the plantation project: lost cities, abandoned train stations, Catholic missions, Spanish fortresses, mass graves, and destroyed forests. The substitution of sugar with soy, too, conjures and multiplies traumatic memories among local workers of the nefarious, supernatural entities who are said to make plantations their abode, including devils, cannibals, and sorcerers who fed off the flesh of labourers, estranged them from their territories, plagued them with disease, and created a general atmosphere of death and of terror (see also Lattas 1993). Such memories are contrasted by the Toba to the ecologies that plantations replace, and particularly the Chaco bush, which is associated with health, healing, and resilience, and with devils that, unlike those inhabiting monocrops, have long entertained reciprocal and life-sustaining relations with humans as guardians and companions. p. 31

The more-than-human distribution of plantation-driven disease, death, and violence constitutes a central motif not only in ethnographic accounts of monocrop lifeworlds but also in literary works, representations, and critiques. Particularly prominent here is the vast corpus of literature stemming from the Caribbean, a region whose history, culture, politics, ecology, and economy have been profoundly shaped by the territory-claiming nature of colonial sugarcane plantations (Benítez-Rojo 1996; Trouillot 2002; Loichot 2013; Hiddleston 2025). For instance, stories and poems authored by Martinican author, poet, and founder of the Négritude movement in Francophone literature, Aimé Césaire (2001), convey a disturbing confusion between human bodies and plantation machinery, with landscapes transformed by intensive industrialization bearing the same physical marks of disease, torture, suffering, starvation, and attrition as the flesh of the labourers forced to sustain them with their blood and their sweat, and that of the environments and species sacrificed in plantations' wake. Similarly, the Martinican novelist and pioneer of the créolité movement, Patrick Chamoiseau (1994), writes of Black workers' bodies mirroring the disposability of landscapes depleted and disfigured to make way for sugarcane. In his vivid descriptions, the mouths of the starved lead them to the cannibalistic mouth of plantations as forest-demolishing, crop-displacing, and labour-sucking machines that swallow the famished whole –like the mouth of the inferno. In both instances, uncanny resonances arise between the rampant ecological defacements produced by plantation proliferation and the facelessness of enslaved Black people themselves, as fungible matter deemed sub-human under rapacious White imperial-capitalist regimes. p. 31–32

Indigenous responses to oil palm expansion across the tropics further communicate the cultural, spiritual, and affective significance of radical ecosystemic shifts resulting from the conversion of biodiverse forests to monocrop plantations. In the Bajo Atrato region of Colombia, for example, forest clearing to make way for oil palm developments is associated by Indigenous and Afro-Colombian communities with an unprecedented explosion in poisonous snake populations, whose bites are notoriously difficult to treat with herbal medicines normally used by traditional curers, and whose threat to human bodies is understood to result from the contamination of the landscape by oil palm (Ruiz-Serna p. 32

2023). The plantation-driven proliferation of lethal creatures such as snakes is compounded by a growing scarcity of native game animals, which communities interpret as symptomatic of a breakdown in relations of respect, reverence, and reciprocity with protective spirit masters who are angered by the relentless deforestation unfolding across the land and threatening native animals. In both instances, damage to the territory as a victim of violence works hand in hand with damage to ecosystems and human and non-human bodies, the collective healing of which is at once urgent and inchoate.

p. 32–33 In Southeast Asia, where oil palm was first cultivated in the form of large-scale monocrops, Indigenous communities often associate plantation developments with experiences of loss, estrangement, regret, longing, and absence that crystallize in affective and cultural terms how ecological declines and disappearances intersect with human wellbeing. Among the Batek of Peninsular Malaysia, for instance, the obliteration of traditionally cultivated and harvested native forest fruits, tubers, and shoots conjures what the Batek call haʔip, or a deep feeling of yearning, love, and nostalgia for the forest's now-ruptured cycles of growth, senescence, and rebirth, and the erasure of nourishing multispecies pasts once embedded in the living landscape (Rudge 2024). The deeply sensory dimensions of haʔip are conveyed in prevalent local characterizations of plantation extinctions through the landscape's changing smells, sights, and sounds: the red sticky substances that indicate blood and death, the unfamiliar screaming of invisible ghosts, and the unpleasant scents of decay that stunt the ripening of fruits. Lived in and through the immediacy of the body, haʔip also takes on an anticipatory or speculative dimension, as people envision or predict potential losses in the future, wherein remaining patches of forest may no longer exist nor be able to offer refuge, sustenance, memories, and knowledges to Batek generations to come.

p. 33–34 In West Papua, where over one million hectares of forest have been razed for oil palm in the last decade, Marind communities with whom I work describe ecological loss and destruction as producing a form of spatiotemporal and bodily alienation that is experienced by humans and non-humans alike (Chao 2024). This alienation manifests in people's incapacity to recognize, relate to, and orient themselves against the regimented uniformity of agribusiness landscapes that are stripped of all spatial and temporal markers, in the disturbing stillness of monocrops as spaces of loneliness and solitude, and in the violent substitution of lively forest bioacoustics with deathly plantation sonics, as singing birds, chirruping insects, and grunting cassowaries give way to roaring bulldozers, gnawing chainsaws, and rumbling trucks. Like the Batek, Marind associate plantation developments with the severance of intimate relations with kindred native plants and animals (amai) that they share common descent with from ancestral spirits (dema) and after whom their clans are named. They also speak of plantations as erasing the spatially embedded past by destroying the sacred sites that were created by ancestral spirits in time immemorial, polluting the patterned bark of sago palms wherein past droughts and fires are imprinted, and depleting populations of wild boars and cassowaries in whose fur and feathers once shimmered the memories of human and more-than-human relatives, both living and deceased (Chao 2022a; Chao 2024). By imposing their mono-time of homogeneous growth on the formerly polytemporal rhythms and overlapping histories of the forest, plantations, as spaces of alienation and disconnection, produce for Marind an out-of-joint present ‑one whose literal and figurative horizonlessness thwarts the possibility of imagining and enacting meaningful, multispecies futures.

3.3 Conservation and Sustainability

p. 34 In recent years, growing awareness of the adverse ecological impacts of plantations ‑ in large part driven by civil society protests within the broader context of the global climate crisis ‑has given rise to a spate of corporate and state initiatives intended to enhance the environmental sustainability of agribusiness projects. These include conservation areas established to offset the ecosystemic damages caused by monocultures, national parks created to protect critically endangered species, organic and fairtrade production systems, and Corporate Social Responsibility protocols, principles, and criteria that integrate socio-environmental concerns into business operations (Colchester, Chao & Jiwan 2012). However, conservation and sustainability initiatives, while arguably laudable in their intentions, are often designed and implemented by scientists, technocrats, and politicians without sufficient consideration for the complex social, historical, and spiritual relations that people on the ground entertain with local landscapes and their constitutive lifeforms.

Take the case of elephant conservation in the northeastern Indian state of Assam, home to the country's largest area of tea plantations and the source of over half of national tea production. Assamese landscapes are shared through intimate and fraught relations of co-presence between humans and elephants, who are worshipped as powerful gods but also feared for their regular and, at times, lethal attacks on human settlements and agriculture (Keil 2025). Far from benign, human-elephant entanglements involve an ongoingly negotiated exchange of needs, mobilities, and survival that is rife with interspecies tension, vulnerability, and friction (Münster 2014). Such tensions are now exacerbated by the state-endorsed establishment of wildlife corridors as a corrective to the habitat fragmentation resulting from deforestation, tea planting, and associated infrastructural developments (Barua 2024).

p. 34–35 Exemplifying a new model of conservation anchored in principles of landscape connectivity and flow rather than

rigid spatial enclosure, wildlife corridors seek to prevent the isolation of herds and thus improve the genetic variability of overall species populations. On the ground, however, the designation of strips of land as ecological corridors has placed growing pressure on farmers to hand over their land and given rise to grassroots protests that stem from radically different understandings of the factors shaping the mobility of elephants. Many local inhabitants, for instance, see themselves, and not the state or conservation scientists, as the guardians of wildlife. They have, since time immemorial, learned to create safe passages for elephants yet are now labelled as encroachers and face eviction from their ancestral lands in the name of conservation. These communities also emphasize that, in contrast to schematic mappings of animal mobility, elephants are highly non-linear in their patterns of movement across the landscape, taking different trajectories in the manner of interlacing trails rather than unidirectional lines. The planned conversion of tea plantations into wildlife conservation habitats poses a particular threat to Adivasi communities, whose ability to transition to other livelihoods is undermined by their status as marginalized Indigenous Peoples and landless workers. As a result, connectivity conservation, intended to limit the environmental damages of plantation proliferation, may very well transform into yet another strategy for expanding state-corporate governance over land and life ‑one that furthers, in an ecological guise, rather than challenges, the uneven human and more-than-human biopolitics of plantation logics (Barua 2024).

In the neighbouring Indian state of West Bengal, home to the tea-growing city of Darjeeling, the ways in which local workers experience and articulate the transition from chemically sustained monocultures to organic and polycultural tea cultivation speak in a different way to the ambiguous or double-edged effects of visions of environmental sustainability, or Green Growth (Kumpf 2024). On the one hand, this transition has enlivened the landscape through the resurgence of myriad lifeforms that are finding new fodder and futures in organic tea plantations ‑from bugs and fungi to owls, hornbills, and tigers. Encounters with these organisms are a source of wonder, surprise, pleasure, curiosity, and much-welcome distraction for plantation workers, their sensory and affective valences helping to interrupt the monotony and repetitiveness of tea plucking. Of particular significance within this emergent ecology is the tea leaf insect, or tea deva in local parlance, which mimics the shape of tea leaves and has returned to the plantations since pesticide use was abolished. Tea devas are venerated as cherished deities, valued as emblems of ecological renewal, and interpreted as evidence of harmonious landscape dynamics. At the same time, however, their sanctioned protection, along with that of other re-emerging lifeforms, has become yet another (often under-compensated) task imposed upon the daily routine of precarious workers who themselves remain economically marginalized and subject to exploitative labour conditions that have changed little, if at all, since the colonial period (Sen 2017). Experiences of wonder and pleasure work hand in hand with realities of exhaustion, disempowerment, and frustration, as a valued species become elevated to an extra-human position in the name of sustainability, but at the cost of the wellbeing of human labourers who remain relegated to the lowest rung of the plantation hierarchy.

p. 35–36

Conservation schemes to protect the critically endangered orangutan in Indonesian Borneo offer a distinctive yet complementary lens into the challenges of achieving ecological sustainability when it is at odds with local ways of knowing and relating across species lines. An iconic mascot in anti-deforestation and anti-plantation campaigns worldwide, there is perhaps no animal that captures the plight of critically endangered species better than the orangutan. Wild populations on the islands of Borneo and Sumatra have more than halved as a result of deforestation and fires caused by oil palm plantations as well as poaching and illegal wildlife trafficking. The species' slow reproduction rates only exacerbate their risk of extinction, which is predicted to occur within the next ten to twenty years (Meijaard et al. 2011, 2018).

p. 36

On the ground, however, internationally funded conservation schemes, including orangutan rescue, rehabilitation, and adoption programs, are often critiqued by local inhabitants, to whom promises of material gain from various development projects, including oil palm plantations, have yet to materialize, and who still do not enjoy basic facilities such as electricity, running water, health care, and communications. For these economically precarious communities, the vast sums of money, as well as time and effort, invested in salvaging and relocating orangutans and then training them to survive in the wild are a source of bafflement and consternation, as well as sentiments of injustice in a world where, as one villager puts it, 'only the orangutans get a life jacket' (Chua et al. 2021: 370). Some express concerns over the potential risks posed by the planned introduction and release of orangutans to their rice paddy fields, which represent their primary source of subsistence. Many see wildlife conservation projects as yet another manifestation of ongoing asymmetries in power between global and local actors, long-standing conflicts over ownership and responsibility, the marginalization of Indigenous communities and voices, and the erasure of customary rights, or adat.

While orangutans share historical and mythical connections to humans alongside other species such as snakes, crocodiles, and hornbills, they are not deemed by communities to be particularly special, nor are they prioritized within local cosmologies. Often, in fact, orangutans are not described by communities as vulnerable victims but rather as agentive entities who are responsible for their own actions (including, importantly, being rightfully killed if posing a life-threat to humans), and who are traditionally associated with mutual avoidance and antagonism, rather than cultivated proximity or intimacy. These local understandings differ markedly from the global representation of orangutans as a charismatic species, capable of attracting compassion, empathy, and action on the part of their all-too-resemblant

p. 36–37

human cousins. They form the grounds for grassroots critiques of conservation projects that are intended to protect endangered species but that, in the process, exacerbate socio-economic injustices for local inhabitants and impose upon them a duty to care for wildlife that contravenes local notions of multispecies agency, relationality, and responsibility.

p. 37 In the Papuan oil palm sector, another understanding of sustainability-driven environmental initiatives is at play. Implemented by agribusiness corporations as part of a commitment to responsible palm oil production under international certification standards such as the Roundtable on Sustainable Palm Oil, these initiatives require that companies identify, demarcate, monitor, and enhance areas classified as 'High Conservation Value'. This includes forests with high concentrations of rare, threatened, or endangered species, and areas that provide critical ecological services such as watershed protection and soil erosion control (Chao 2019). Yet, despite their purportedly beneficial value, conservation zones are a source of widespread frustration and sorrow for many Indigenous Marind communities. For starters, what distinguishes a high from a low conservation value zone is unclear. Plants and animals, Marind emphasize, depend upon diverse ecologies that are all equally important to their subsistence, reproduction, and growth – swamps, marshlands, mangroves, savannah, forest, and more. While non-humans are restricted to bounded zones of liveability, Marind themselves are now excluded from areas where they once hunted, foraged, and fished, as access to these areas is forbidden to all except company personnel, conservation scientists, and sustainability auditors. Reciprocal relations of care between diverse organisms are thwarted by a stratification-cum-dissection of space and species that runs counter to Marind's characterization of the forest as a realm enlivened by the sympoietic interactions of humans and other-than-humans. In the process, the unjust exclusion of Indigenous communities from the ecologies that sustain them goes hand in hand with the unjust severance of sentient ecologies themselves from the humans who care for and maintain them.

p. 37–38 Species cut off from their relations to humans in the name of environmental conservation are a source of great sadness (kesedihan) for Marind. Most notable among these is the sago palm, a plant of central significance in Marind culture and the source of their staple starch, sago flour (Chao 2021c). In conservation zones, Marind cannot care for sago palms through traditional practices like trimming, selective felling, and transplanting that enhance sago's access to sunlight and nutrients and render their environment more conducive to healthy growth. As selective transplanting cannot take place, palms in conservation zones tend to reproduce sexually rather than vegetatively. In contrast to vegetative reproduction, this ends the life cycle of the parent palm, whose seeds disperse across great distances and germinate far away from their sago kin. These seedlings tend to differ morphologically from their kin, unlike vegetatively propagated suckers that form clonal colonies of similar genetic makeup to their parent. They also exhibit a slower and less regular rate of growth. Separated from each other, sago seeds and their parents are said to become 'dry' (kering), 'lonely' (kesendirian), and 'sad' (sedih) –sentiments that are also experienced and described by Marind themselves as traditional sago carers. Meanwhile, the segregation of palms to conservation zones negates the possibility of creating shared memories and relations between people and plants. As Pius Gebze, a village elder and close friend, explained to me, 'when sago and humans grow apart, they suffer because they no longer share skin or wetness. They can no longer make stories or futures together. This is the violence of conservation'.

3.4 Conclusions

p. 38 The material presented in this section uncovers how plantations are grounded both in the necropolitics of a 'hegemonic, extinction-oriented creed' (Tsing 2011: 19) and in an instrumental biopolitics that requires that anything *not* supportive of cash crop liveliness and lucrativeness either die or be displaced (Hetherington 2019; Besky 2020). The monocultural logic undergirding these conjoined processes, as Indian activist and scholar Vandana Shiva (1993) critiques, is intolerant of alternative systems of cultivation or knowledge, and primarily economistic and anti-pluralistic in its telos. This intolerance includes a common disregard for long-standing Indigenous and local ways of relating to the environment and its more-than-human dwellers, and for the culturally and spiritually shaped ways in which people on the ground experience, interpret, and engage with rapidly transforming landscapes (Kröger 2022). It also comes tethered to an epistemic myopia regarding the ability of non-human life to exert its own influences and effects over and against human-driven projects, as exemplified by the activities of pests, parasites, and other feral organisms who repurpose capitalist natures as homes and hosts (Tomich 2011; Ovstehage 2021).

p. 38–39 Initiatives that seek to enhance the sustainability of the plantation sector, yet that remain premised on top-down, technocratic epistemologies and praxiologies, can all too easily end up replicating the very same hierarchies and exclusions that lie at the heart of the plantation system itself – human/non-human, nature/culture, wild/domesticated, and so forth. This includes conservation practices that are predicated on the belief that biodiversity protection is best achieved by isolating ecosystems from humans, now reframed as sources of disturbance, and that therefore come to perpetuate the colonial paradigm of 'fortress conservation' (Brockington 2002; see also Milne 2022). The result is a form of sustainability that dispossesses and excludes local communities by conjuring natures worth preserving as pristine and untouched zones of 'wilderness', while simultaneously displacing and erasing place-based relationships, meanings, values, and histories (Cronon 1996; Tsing 2003). Such forms of sustainability constitute manifestations

of ‘eco-imperialism’ (Driessen 2010) that further ‘accumulation through (double) dispossession’ (Harvey 2003) by failing to acknowledge the intrinsic engagement of humans with their environment and the encompassment of other-than-human lifeforms within the multispecies society that constitutes ‘nature’ itself. In designating only particular sites and landscapes as worthy of conserving, they elide critical local, relational, and holistic understandings of the landscape as a living whole – including the sometimes unexpected frictions and tensions that shape the material and moral relations of humans to non-humans.

4 Emergence

p. 39–40 The extractions and extinctions traced in the two preceding sections conjure plantations as landscapes that have, over time, proved lively and lethal, anthropocentric and racializing, violent and vulnerable, ruinous and resistant, and coercive and creative. These landscapes have been experienced, understood, and responded to differently by situated human and non-human subjects, thus transforming plantations into sites of emergence –our third analytical entry point into plantation lifeworlds and afterlives. By emergence, I refer to how particular ways of *knowing, being, relating, and representing have come into existence, gained (re)new(ed) prominence, or acquired inflected significance* in response to plantation logics and legacies, and how these processes and practices speak to the more-than-human actors and agencies animating plantation structures, systems, and societies. I begin by considering plantations as laboratories for colonial, postcolonial, and anti-colonial articulations of disease, race, and science. I then examine the diverse meanings, moralities, and materialities emerging from peoples' engagement with plantations as environments where new intimacies with cash crops come into being and as the birthing grounds of contemporary foodways and anti-colonial food sovereignty movements. The section further explores how the racializing and ecocidal force of plantations has spurred the formation of new identities, imaginaries, aspirations, commonings, and belongings that are defined –if not always determined –by shared experiences of vulnerability and resistance to plantation violence. The final section turns to art and aesthetics as politically charged representations, expressions, and critiques of plantations' attritive yet never unchallenged effects and affects, as these are distributed across disparate places, peoples, and periods.

4.1 Illness and Healing

p. 40 The intensively cultivated environment of industrial monocrops, combined with the large-scale and often transnational migration of labourers required to tend to them, conspire to make plantations a particularly potent breeding ground for human and non-human epidemics as well as emergent epistemologies of illness and healing. Indeed, plantations have long acted as informal laboratories for the study of biomedicine and disease ecology, the latter referring to the interplay of pathogens, hosts, and ecosystems that influence the dynamics of infectious diseases, including their transmission, spread, and impact.

The close relationship between human and non-human actors in the making of plantation pathologies is exemplified in colonial discourses of 'plantation hygiene', which were premised on the belief that diseases emanated from unhealthy elements unearthed from the soil through land clearing, leading to strict directives for keeping plantation landscapes as organized as possible –for instance, through building, drainage, and ditching –to sustain the health of crops, animals, and the enslaved (Aso 2018). Yet even as their bodies were harnessed to further biomedical knowledge production, enslaved and otherwise exploited plantation workers often toiled amidst deplorable working and living conditions, their wellbeing sustained through selective biomedical interventions to the ends of enhancing their productivity (and, in the case of women, their reproductive abilities) as sources of labour. The bodies of enslaved men and women in colonial plantations were also routinely exploited as testing grounds for non-consensual surgical experiments and sources of medical specimens within biomedical research that supported proslavery ideology and White supremacy. In a mutually reinforcing process, these bodies became the basis from which early nineteenth-century scientists, physicians, and polygeneticists elaborated models of civilization that posited racially differential capacities to feel pain and divergent neurological functions, immunities, sexualities, and moralities based on the putatively separate origins of distinct races (Eersel 2021).

p. 40–41 The deep and disturbing connections between biomedical advances and plantation regimes are perhaps most starkly exemplified in the case of gynaecology, a field where early surgical trials were conducted on enslaved Black women working in cotton, sugarcane, and tobacco plantations, who were leased to American scientists for biomedical experimentation (Owens 2017). This includes trials undertaken in the 1840s by Dr. James Marion Sims, the so-called 'father of modern gynaecology', whose role in furthering subsequent advances in the field led to his election as president of the American Medical Association in 1875 and the American Gynaecological Society in 1879. Sims's pioneering success in repairing vesico-vaginal fistulae came off the back of countless crude experiments performed on Black female plantation workers (thirty surgeries over five years, in the case of one subject), without any form of anaesthesia. In these and many other cases, the treatment of enslaved women, while motivated by therapeutic ends, was more often than not driven by the desire to protect the economic interests of slave owners through maintaining sound female reproductive bodies. They also offered budding physicians the opportunity to practice and perfect their surgical skills on systematically objectified and easily accessible (if also entirely non-consensual) Black bodies.

p. 41 Racially inflected biomedical ideologies, birthed in the nineteenth century, were to perdure in different guises throughout later decades. In the Liberian rubber plantation sector, they manifested as experiments conducted by Western parasitologists in the 1950s, which involved intentionally infecting Liberian staff and residents with a strand of malaria (Plasmodium vivax) that was not present in the country, to determine if Black people possessed an innate immunity to the disease (Mitman 2021). In the post-plantation landscapes of the American South, they found embodiment

in the Tuskegee Study, a project since recognized as the more concrete symbol of scientific and medical injustice in the United States (Rusert 2019). Conducted by the Public Health Service for four decades spanning 1932 to 1972, as part of the US's quest for global biomedical dominance, the Tuskegee Study involved hundreds of African-American men being unknowingly observed and used as subjects to track the natural progression of untreated syphilis, the diagnosis of which was withheld from the subjects alongside the treatment with penicillin. Central to the clinical research resulting from the study was the representation of African-American men as non-modern, timeless people connected to a primitive and enclosed landscape – a characterisation that bore clear roots in intergenerational histories of enslavement, the carceral topographies of colonial plantations, and the racially segregated debt peonage system under which many of the research subjects were, at the time, still working.

But epistemologies of illness and healing borne from plantation regimes were not determined solely by Anglo-European agents and agendas. Rather, they became fertile, if contested, terrains for the re-emergence of transplanted traditional ethnomedicines on the one hand, and for the cross-pollination of different understandings of biomedicine and its relationship to the natural and spiritual world on the other. In the antebellum US South, for instance, enslaved Black communities of Igbo, Yoruba, Bambara, Kongo, and other heritages resisted the normativity of medical abuse and racist scientific theories by developing and transmitting protective and healing therapies that found root in a relational ethos situating individual health in relation to community-wide harmony, ancestral wisdom, divine revelation, the ability to hear the spirit of medicinal plants, and the intimate connection between gods, humans, and the environment (Fett 2002). Central to these therapies were the familiar medicinal plants that enslaved peoples brought with them across the Transatlantic Passage – from liquorice to treat coughs and fevers, to okra leaves as poultices, senna as a laxative, and kola beads for stomach aches – as well as the Indigenous plants that plantation workers cultivated or harvested from forests, swamps, and wetlands at the edges of plantations – from dogwood root to treat fever, to jimsonweed for dropsy, Jerusalem oak for asthma, and capsicum for indigestion. Delivered primarily by women in their roles as midwives, herbalists, and root doctors, these therapies involved creative adaptations of long-standing African healing traditions to the ecologies and vicissitudes of the New World, an intimate knowledge of the plantation landscape, as well as mutual borrowings from Native American medicine and European science, which together profoundly shaped contemporary Black biomedical cultures, including Brazilian Candomblé, Haitian Voodoo, and North American hoodoo.

p. 41–42

Herbal healing as a form of resistance was also at play in the cotton, coffee, timber, and sugarcane plantations of Suriname, where enslaved Africans cultivated a highly intricate understanding of the landscape and its vegetal lifeforms through intertwined processes of trial and error, obtaining information from Indigenous communities, and identifying phenotypic similarities between New and Old World plants (Aaron-Denz 2021). Prominent among these were korsuwiwiri, which was used to treat fever; usuwe, to ward off insects; tobacco, to counter poisonous snake bites; and redi katun, to treat eye and ear inflammations. The mutually informing nature of explanatory frameworks for disease and illness has more recently been evidenced in biomedical research conducted in Suriname on leprosy, a highly contagious condition that was particularly rampant among concentrated communities of enslaved and indentured plantation workers (Menke et al. 2021). Traditional Afro-Surinamese beliefs that leprosy is caused by the breaking of a taboo on the consumption of totemic animals such as the iguana, tortoise, snake, and Kapasi are now being backed by growing scientific evidence that specific wild animals – notably armadillos and Kapasi – do appear to carry leprosy bacteria and may play a significant role in leprosy transmission. Rather than conceptualizing these explanations as mutually exclusive, many Surinamese reportedly understand leprosy to be both a bacterium-caused infectious disease *and* a consequence of supernatural, punitive intervention resulting from the wrongful killing and consumption of revered animals. This conjures how plantation pathologies, together with their diagnoses and treatments, are shaped by multidirectional processes of trans-culturation (Ortiz 1947) wherein new or modified ways of knowing and healing emerge from the cross-pollination of differently situated and rooted aetiologies and epistemologies of disease.

p. 42–43

4.2 Vegetal Ontologies

A second modality of emergence within plantation lifeworlds centres on communities' changing material, moral, affective, and cultural relationships to cash crops that have been introduced into newly established agro-industrial geographies or that are endemic to these landscapes but have been repurposed for large-scale monocrop cultivation. Attending to these emergent entanglements, together with their attendant allyships and animosities, uncovers how plants and people have co-produced each other through historical and contemporary processes that crosscut plantation, counter-plantation, and post-plantation lifeworlds and legacies, in a multispecies dynamic that Andrews Flachs et al. (2024) term 'plant-anthropo-genesis'.

p. 43

At one end of the plant-human affective spectrum is antagonism. This response comes to light in Paraguay, the world's fourth-largest exporter of soybeans, where activists set a legal precedent in bringing soy beans before the courts as legitimate persons and actors, rather than solely as instruments of human agency and structural forces (Hetherington 2013). Referred to as 'soy that kills' (la soja mata) in judicial proceedings and in popular slogans, the plant was understood and represented as having a force of its own – one that demanded recognition within Paraguayan human

and more-than-human political processes and that expanded the scope of responsibility to include plants as perpetrators of harm, drivers of injustice, and causes for redress. At the other end of the human-vegetal spectrum is intimacy. Exemplifying this are the fictive and actual forms of kinship that women plantation workers entertain with tea plants in the Indian state of Darjeeling (Besky 2013: 59–87). These women describe tea-bushes as their 'children' and themselves as tea-bushes' 'mothers' and 'grandmothers', blending ideas of social reproduction with visions of multispecies relatedness (see Figure 4). Tea plants are cherished as more-than-human kin, even as the labour they demand of humans is also recognized by women as hugely demanding and often exhausting, and even as women nurture these plants within a broader context of gendered control, wherein monopoly over plantation operations is exercised by primarily male plantation managers.



Figure 4 An Indian female worker plucking fresh leaves in a tea plantation. Image credit: Pradeep Weragoda.

p. 44–45

In other contexts, plantation cash crops come tethered to the promise of expanded worlds, relationships, and connections. In Papua New Guinea, for instance, Gimi communities contrast the more-than-human socialities afforded by endemic, small-scale cultivated sweet potato with the lack of cultural value of introduced, industrially produced coffee, at the same time as they talk of coffee (rather than sweet potato) as enabling them to establish new relations outside Gimi society with global markets, traders, and consumers (West 2012). In the process, coffee as commodity and imaginary comes to stand in for being modern, participating in development and social change, and expanding one's kinship relations beyond one's own community. Across the New Guinea border in West Papua, meanwhile the ontology of cash crops is characterized primarily by ambivalence and ambiguity (Chao 2022a; Chao 2020). While Indigenous Marind communities do associate oil palm with the possibility of an expanded social world and connections, they primarily describe this introduced cash crop as 'abu-abu' (meaning both grey and uncertain) and often express resentment, anger, fear, and hatred towards it because it furthers their own displacement, disempowerment, and dispossession. These affective responses, however, work hand in hand with widespread sentiments of pity, compassion, and empathy for a plant that is itself subjected to human, institutional, biological, and technological violence and thus deprived of its 'freedom' (kebebasan) to grow autonomously. Compounded with pity and empathy are curiosity, wonder, and speculation, as Marind, now forced to live in oil palm's company, muse over the plant's own native homelands, the possibilities of forging less violent relationships with it, and the distant landscapes and lifeforms that oil palm connects them to across continents and oceans, in the myriad places where its produce –palm oil –travels to.

p. 45

Other forms of human-plant emergence come to the fore in today's post- and counter-plantation landscapes, including interspecies relatings that transcend or resignify native-invasive species distinctions. Across Latin America, for instance, banana trees, first introduced by European colonizers in the sixteenth century and then industrially cultivated in the nineteenth century by an exploited and racialized labour force have since been creatively repurposed by Indigenous Peoples as sources of food, cooking, medicine, and shelter-building materials (Hernandez 2022). The plant is not native to the continent –but neither is it seen as invasive by local inhabitants, who instead have incorporated it within their cultures, diets, and kinships with the land. A similarly emergent human-plant allyship is at play in Brazil, where introduced oil palms (dendezeiros) that now grow in feral form are valued by Indigenous Pataxó communities for

their ability to co-exist symbiotically with native birds, rodents, reptiles, and humans, and where Afro-descendants of formerly enslaved plantation workers have successfully maintained oil palm polycultures that sustain local alimentary and spiritual demands while resisting the advance of modern oil palm monocultures (Cardoso 2016; Watkins 2021). In both instances, plantation crops are recognized by local people as living in place, interacting with others, and having a right to exist and flourish in ways that defy the homogenizing and simplifying force of the plantation system.

In the African island-nation of São Tomé, where colonial sugar and coffee plantations were established in the sixteenth and nineteenth centuries respectively, it is not with cash crops themselves that workers established new intimacies, but rather with a plant that thrives at the plantations' unruly margins: cannabis. As a source of pleasure, leisure, and subversive power, cannabis and its biodiverse forest ecologies came to epitomize for workers the possibility of nurturing counter-plantation lifeworlds in solidarity with vegetal beings who helped them heal themselves, reach enlightened reasoning, withstand work, connect to metaphysical forces, and find moments of fun and solace amidst the violence and tedium of plantation labour (Macedo 2024). A comparable dynamic underlies the provision grounds, or plots, that enslaved workers cultivated in the Americas and the Caribbean and that were rooted in ancient and new knowledges of and relationships with plants as food and as kin – from millet, sorghum, and coffee, to rice, okra, and watermelon (Carney & Rosomoff 2011). These subaltern, more-than-human contact zones would eventually produce many of the foods and cuisines of the American diet, their legacy as counter-plantation ecologies manifesting vividly in the present-day proliferation of grassroots Black agricultural resistance and food sovereignty movements across the United States, many of which trace their roots back to the plantation plot as a locus for the nurturing of oppositional modes of Black life and multispecies alliances (Carney 2001; White 2018; Davis et al. 2019). p. 45–46

Finally, monocrops have also been the source of emergent ecologies whose protagonists and possibilities extend well beyond those imposed or imagined by plantation logics. A striking example of this is found in contemporary Japan, where the ruins of former timber plantations have become the flourishing grounds of the matsutake mushroom, the world's most expensive fungi and a species that thrives best in disturbed landscapes. Exemplifying what Anna Tsing (2015: viii) terms 'third natures', matsutake mushrooms both depend upon and subvert the deadliness of industrial plantation landscapes, pointing to other kinds of collaborative survivals in the face of environmental destruction. These organisms offer their pickers a sense of freedom, community, and play that operates alongside an awareness that the landscape is haunted by long-disappeared plants, animals, and ancestors who nonetheless remain alive in ghostly forms. In doing so, matsutake mushrooms bring us to reimagine possibilities for interspecies collaboration, coexistence, and companionship in the face of plantation-induced precarities that sit within broader processes of environmental crisis and planetary decline and that demand that we, like matsutake, learn to live together and otherwise in a world of ruins. p. 46

4.3 Identities and Imaginaries

A third aspect of emergence concerns the identity formations, social movements, cultural loci, and collective imaginaries that have arisen in response to plantation logic and its attempts to classify, hierarchize, and therefore better discipline and exploit, human and non-human lifeforms. In the context of the Trans-Atlantic slave trade and colonial plantation systems, these dynamics sit within a broader process that Paul Gilroy (1995) terms the 'Black Atlantic culture', or a culture that transcends ethnicity and nationality in its composition, reach, and dynamics. Neither specifically African, American, Caribbean, nor British, Black Atlantic culture encompasses all these cultural currents at once and differently, generating a pan-Atlantic nationalism that is not premised on a spatially bounded nation but rather rooted in shared experiences of plantation violence and resistance.

Central to the emergence of Black Atlantic culture was the practice of marronage, which refers to the flight of enslaved individuals and groups from plantation control that, in turn, produced 'maroon societies' across the Caribbean, Latin America, and the United States (Price 1979; Thompson 2006). Most notable among maroon societies are the Palmares in Brazil, the Saramakas and Ndjukas in Suriname, the San Basilio in Colombia, the Esmeraldes in Ecuador, the Le Maniel on the border of Haiti and the Dominican Republic, and the Leewards and Windwards in Jamaica. Marronage varied in scale from 'petit marronage', where individuals or groups escaped the plantations for days or weeks and then returned, to 'grand marronage', where people fled the plantations permanently. The societies that emerged as a result of these movements ranged from small, short-lived bands to powerful, enduring states encompassing thousands of members. In Suriname, for instance, the maroon population represented around 10 per cent of the enslaved population in 1786, and in Venezuela, around half by the end of the eighteenth century. While states did not grant them legal recognition at the time of their formation, these communities were to constitute the first independent polities from European colonial rule. p. 46–47

Closely tied to marronage in the emergence of Black Atlantic culture was creolization, which arose as African enslaved peoples and their descendants synthesized cultural elements from their native homelands with ideas and practices from European colonists. This process of selective creation and cultural struggle produced 'creole societies' whose patterns of existence, modes of thought, codes of behaviour, and ability to endure through innovation and adaptation were enabled in large part through the aforementioned nurturing of provision grounds on plantation peripheries (Trouil- p. 47

lot 2002). It was in and with these nourishing, more-than-human ecologies that creole communities were able to renew practices of cooperative labour, social time, multispecies kinship, song and dance, and conversation and care that were denied to them under the dehumanizing violence of plantation regimes.

p. 47–48 While arguably less examined within plantation-focused scholarship, agroindustrial systems have also inflected pan-regional configurations of 'whiteness', including in ways that invoke multispecies histories, geographies, and intimacies. In Hawai'i, for example, the category of Portogee, pocho, pukiki, or simply Portuguese finds its roots in the arrival and settlement of sugar plantation contract labourers from the Atlantic islands of Madeira and the Azores in the late nineteenth and early twentieth centuries (Bastos 2024). Present-day descendants of these workers situate their social and historical positionalities in Hawai'i in relation to these intergenerational histories of plantation migration and labour. But they also frame their collective identity in association with the plant of sugarcane who first brought their ancestors to the island and who has come to embody by extension a distant, quasi-mythical, and beloved place of origin. At the same time, the association of Portuguese populations with plantation labour has historically positioned the diaspora in Hawai'i as 'off-white' in terms of skin and class ⁻Caucasian, but not 'White' in the privileged manner of Anglo plantation owners, Protestant missionaries, or their respective descendants.

p. 48 Finally, the emergence of social groups, collective identities, and shared imaginaries resulting from plantation regimes encompasses expressions and enactments of Indigeneity that have been hybridized, contested, reclaimed, and heterogenized across time and space. In the colonial American South, for instance, anti-plantation revolts and rebellions in the early nineteenth century were led by the Black Seminoles, a people formed through the inter-ethnic allyship of Indigenous communities with enslaved Africans escaping bondage and welcomed by Indigenous groups who, in joining forces, initiated the most coordinated uprising of Indigenous Peoples and Black people in US history during the Second Seminole War (1835⁻1842) (Covington 2001; Hatch 2012; Porter 2013). The historical emergence of Seminole identity and defiance through inter-ethnic solidarities stands in contrast to post-plantation Hawai'i, which is characterized by a far more vexed landscape of identity-formation and claims to rootedness. Here, dominant articulations of Indigeneity stemming from East Asian Americans whose ancestors were imported as workers to toil in sugarcane monocrops produce an understanding of belonging that is anchored primarily in plantation labour rather than ancestral connection. While these articulations center the role of work in the production of plantation and post-plantation identifies, they simultaneously obscure the egregious exploitation of Indigenous Kānaka Maoli under the plantation regime and the broader settler-colonial order that endures in Hawai'i as American-occupied territory (Day & Hogue 2019).

p. 48–49 Indigeneity is also and differently at play in South Africa, where plantation developments are reconfiguring understandings of and claims to nativity among San People, or Bushmen, and white South Africans, or Afrikaans. For example, the transformation of hoodia ⁻a succulent plant traditionally used by Indigenous San as an appetite suppressant, energy enhancer, and breastfeeding aid ⁻into a popular product of the global diet industry has been accompanied by processes of scientific research, patent ownership, contractual benefit-sharing mechanisms, and bioprospecting legislation that have created opportunities for San People to articulate their claims to ownership, intellectual property, and self-determination (Foster 2017). The case of rooibos, meanwhile, illustrates how the cultivation of a crop celebrated as 'ecologically indigenous' by people of 'white' African Indigeneity (in contrast to the 'coloureds') conjures new, plant-based expressions of belonging and exclusion (Ives 2017). Instead of grounding their sense of rootedness to place, rooibos cultivators find possibilities in and metonymic identification with rooibos ⁻an indigenous plant ⁻and with rooibos farming, thus articulating not only a claim to the past but also to the future, achieved through engagement in the lucrative rooibos industry.

p. 49 At the same time, Indigenous emergences prompted by plantation proliferation are often diverse and conflicted (Roberts 2019). In Papua New Guinea, for instance, struggles over land, resources, and development are increasingly polarizing migrants and landowners, resulting in episodic tensions and violence across the palm oil sector (Koczberski & Curry 2004). Despite this, many customary landowners have been successful in maintaining an indigenous, though partly modified, system of land tenure that remains grounded in place-based social practices and values imbued with indigenous moralities (Curry & Koczberski 2009; Tammisto 2019). Land retains its traditional meaning as a source of history, identity, rootedness, and sacrality, but is now also understood and worked upon by Indigenous communities as resource and commodity (Tammisto 2024). As such, disagreements over the promise and perils of monocrop development are often disagreements over how to engage with, and benefit from, the plantation industry, while also preserving or establishing socially productive relations, both with other people, and with the land itself.

4.4 Art and Aesthetics

Our final foray into Plantationocenic emergence centres on art and aesthetics as political critique, creative resistance, social protest, and practice of repair, fashioned and transmitted by those subject to plantation regimes and their conjoined racial and ecological violence. In the realm of music, the connections between artistic expression and plantation histories are perhaps nowhere as palpable as in the blues, a genre that emerged in the post-emancipation, segregated American South, and that is rooted in the field hollers, work songs, and ballads composed and improvised by enslaved plantation workers (Bindas 1992). Blues songs took the form of rhythmic call-and-response chants and accompanied the harsh yet tedious monotony of plantation labour ⁻and later, that of forced penal labour on Black prison farms. While blues songs were often encouraged by plantation owners for their positive effects on the stamina and productivity of the enslaved at work, their lyrics often contained concealed or cryptic critiques, condemnations, and even mockery of the plantation system and its White agents, in what might be described as a form of lyrical resistance. A central influence on African-American jazz and related musical currents, these songs represent one facet of the broader phenomenon that Clyde Woods (1998: 17) terms 'blues epistemology', or the Black revolutionary consciousness vested in the deconstruction of plantation relations.

Invocations of workers' native lands, plants, and communities, as well as references to non-humans of the New World, were prevalent within plantation blues across the Black Atlantic, which often entailed a synthesis of African and European musical forms (Nunn 2016). In the early twentieth-century Mississippi Delta, for example, Black tenant farmers created a vast body of so-called 'boll weevil songs' that centred on a deadly pest that would eventually devastate the cotton industry. The lyrics of these songs conjured the boll weevil as a mobile, defiant, and resilient organism in constant search of a home, its invasive spread threatening the cotton sector at the same time as it mirrored the spirit of tenants whose frequent migrations were, not unlike those of the weevil, often driven by the pursuit of better opportunities (Giesen 2011). In this respect, boll weevil songs were as much about learning to live with cotton parasites as they were about Black tenants' precarious positioning within cotton culture itself. p. 50

A different multispecies edge to plantation-inspired, emergent musical forms manifests in the 'Black Pacific' (Webb-Gannon & Webb 2018; Webb & Webb-Gannon 2020). While Melanesians have long sought to distance themselves from the 'natural' world due to colonial constructs that instrumentalized their Blackness to relegate them to a putative primitiveness, many lyrics from the region celebrate the more-than-human realm alternately as an Edenic, tropical paradise, as God's gift to Melanesians and to the world, or as a source of wealth and identity plundered through extractive capitalism and colonialism.

A powerful example of music as multispecies protest in the Black Pacific is the tradition of mourning songs practiced by Indigenous West Papuans on the oil palm frontier (Chao 2023b). Performed by men, women, and children near monocrops, roads, and razed forests, mourning songs, which began to spread following the arrival of palm, condemn and grieve the indiscriminate killing of native organisms due to monocrop developments while celebrating their meaningful pasts, histories, and kinships with local clans. Much like pidginized blues songs once acted as a veiled medium to communicate secret messages among the enslaved, so too Papuans describe multispecies mourning songs as a form of creative resistance and collective defiance ⁻one that pushes against the disenchantment of plantation modernity through an insistence on storying and singing the lifeways and deathways of more-than-human beings subject to plantations' ecocidal effects.

Alongside music, the reconceptualization of human subjecthood through the animacy of non-human forces runs through literary works focused upon or inspired by plantation lifeworlds and afterlives. In the United States, this is visible in the vast body of poems, novels, and memoirs produced by African-American descendants of enslaved plantation workers across the twentieth century and beyond. As expressions of the 'Black radical imagination' (Kelley 2002), these oeuvres collectively constitute what Christina Sharpe (2016) calls 'wake work', or counter-hegemonic sites of Black artistic production, resistance, consciousness, and possibility that address the 'monstrous intimacies' of slavery and their repetition in the form of post-slavery Black subjectivity (see also Sharpe 2010). p. 50–51

Paul Laurence Dunbar's plantation poems and W. E. B. Du Bois's *The Quest of the Silver Fleece* offer powerful examples of how Black intellectuals have reimagined the plantation landscape as animate matter, capable of storing memory, articulating pain, and enabling defiance (McInnis 2016; Child 2019). In Du Bois's novel, the 'black soil' of former plantation landscapes becomes a fecund ground for radical freedom and independence, its capacity to anchor collective autonomy inseparable from the recognition of Black landownership. Darkness, likewise, operates as a more-than-human force within the work, as exemplified by the strange music that rises from murky swamps and Black bodies. Resignified as a quality rather than a deficiency, darkness in the novel further acquires tactical value in enabling black invisibility during mob attacks, exemplifying a repurposing of racial hypervisibility into a resource of protection. In both respects, soil and darkness are positioned as active collaborators in Black life-making, charged with generative and resistant potential. Dunbar's poetry, meanwhile, turns to haunted landscapes in which trees and soils register the afterlives of plantation violence. One example is 'The Haunted Oak' (1903), which stages a dialogue with a lynching tree whose bark absorbs the agony of a murdered black man and whose testimony survives where human voices are p. 51

forcefully muted. The tree's spectral utterances collapse distinctions between subject and object and human and vegetal, indicting regimes of ownership that once fused White control of over bodies, labour, and land. The non-human emerges as both archive and witness, bearing the imprint of racial violence. Foregrounding soil, darkness, and trees as co-actors, Du Bois and Dunbar's works refuse to confine plantations to a histories of subjugation. Instead, these works recast plantations and their ecological and material residues as sites of agropolitical possibility in the making of a Black subjecthood that is grounded in the embrace of cross-species entanglements and the recognition of the political vitality of the non-human.

p. 51–52 Non-human actors and forces figure in plantation-inspired literary works produced across many other world geographies beyond the Black Atlantic, where monocrops have also radically reshaped multispecies pasts, presents, and futures. In the Latin American context, for instance, Argentinian author Samanta Schweblin's award-winning novel *Fever Dream* (2017, originally published in 2014 in Spanish as *Distancia de rescate*) deploys magical realism and ecohorror to story the toxic contamination of water bodies from chemical pesticides used in soy plantations and its subsequent connections with birth defects, neurological imbalances, and fertility problems, in a haunting tale of anthropogenic monstrosity replete with deformed beings, poisoned animals, dystopic futures, and collapsing ecologies (Sarkar 2023). Fijian authors Totaram Sanadhya's 'The Story of the Haunted Line' (1922) and Subramani's 'The Fantasy Eaters' (1988) lean into sugarcane plantations in the gothic mode, bringing to life the slow violence of food insecurity caused by monocrop developments, the threats of mosquito-borne diseases, the exhaustion of human and environmental fertility, the disciplining of life into cheap energy, the multispecies labour regimes of the monocrop, and the ghostly social, economic, and environmental legacies of plantation lifeworlds (Vandertop 2020). Indian author and film-maker Satyajit Ray, meanwhile, delves into the haunting horrors of colonial plantations in his short story 'Indigo' (1968), which blurs the line between the real and supernatural through the recounting of a young Indian man who is possessed by the spirit and body of a nineteenth-century English plantation owner and tax lord while overnighting in an abandoned bungalow in the indigo-producing Bengalese district of Birbhum. In Amitav Ghosh's *The Nutmeg Curse* (2021), the parable offers a powerful stylistic form for uncovering historical continuities and connections between climate change and the decimation of indigenous forests and populations to establish nutmeg plantations for the European market in the Dutch-occupied Banda Islands. Both colonial violence and contemporary planetary crises, Ghosh demonstrates, remain rooted in an imperial mindset wherein the Earth exists solely as a resource for profit, and its exploitation continues to contribute to environmental degradation and social inequality.

p. 52–53 Beyond the realm of literature, creative modes of storying plantations are now also seeping into the realm of scholarly writing, as exemplified by anthropologist Michael Taussig's *Palma Africana* (2018), which speaks to the ravaging impacts of monocrop oil palm plantations in Colombia (Chao 2019). Described by Taussig as 'animistic writing' or 'ethnographic (sur)realism', *Palma Africana* is populated by a panoply of human and non-human actors: from paramilitaries, peasants, and activists to cows, cocaine, and cash crops that entangle with swamps, migrant camps, polluted rivers, braying donkeys, and ancient Indian fields. 'Thought-images' in the form of introspective musings, hand-drawn pictures, and uncaptioned photographs, delve into the sounds and atmospheres of industrial monocrops, desolate oil palm towns, and dying forests. The amorphous narrative of the text jolts the reader from the empirical to the reflexive, the violent to the mundane, and the human to the other-than-human. In embracing the magic and madness of plantation violence, *Palma Africana* counters what Taussig (2015) terms 'agribusiness writing', or a form of writing that, like the plantation system itself, is stripped of its capacity to provoke surprise and wonder and thus entrenches the dangerous fiction of human mastery over multispecies realities.

p. 53 Finally, numerous examples of plantation and counter-plantation imaginaries can be found in the realm of the visual and performative arts. In the American context, one might think of Brooklyn-based Jamaican photographer Radcliffe Roye's award-winning 2016 photograph, 'Don't Shoot!', which portrays a young man of colour holding his hands up in a field of white cotton, and which was taken near Clarksdale, Mississippi, soon after the police killings of African-American teenagers Michael Brown and Eric Garner. The image is one of 4,000 photos linking the enduring social legacies of plantations in the American South, including ongoing Black American experiences of racial terror, segregation, surveillance, and policing (Aikens et al. 2019). Or, one might envision African-American painter Kara Walker's Brooklyn 2014 installation, 'A Subtlety: or the Marvelous Sugar Baby', which features a colossal nude sphinx sculpted from foam and coated with refined sugar, surrounded by smaller figures of children cast in brown sugar and molasses who appear to melt into the floor of the former sugar refinery where the installation is located. The work evokes the intertwined histories of sugar plantations, capitalism, enslavement, labour, violence, racism, and desire, exposing how White consumption depended on plantation commodities and Black suffering.

Further south in Latin America, contemporary Cuban artists have harnessed the conceptual, political, and aesthetic power of intermediality, or the use of multi-layered mediums and techniques, to represent the symbolic and social afterlives of sugar plantations (Sklodowska 2022). For instance, Tania Bruguera's performance, *El Peso de la Culpa* (1997–99, *The Burden of Guilt*), in which she ate soil mixed with salt water for forty-five minutes, evokes Indigenous Taíno resistance under Spanish colonization, which often involved the symbolic ingestion of soil, while also conjuring the horrific bodily punishment and iron-mask muzzling of enslaved plantation workers who sought to suicide by

consuming dirt. The dehumanization of Black people turned instruments of colonial capitalism, meanwhile, is vividly captured in the photography of Reinaldo Echemendía Cid, and particularly in *Con-tracción* (2015, *Con-Traction*). In this striking image, a stretched out, naked, muscular Afro-Cuban man acts as a coupling rod holding together the wheels of an antique train, his power and labour standing in for the locomotive's engines and his energy metabolized to accelerate production. The photograph speaks both to the herculean strength of the enslaved and to their reduction to expendable and interchangeable cogs, instruments, and automatons within the profit-making plantation machine.

Another lens into plantation lifeworlds is offered by those visual artworks and performances that centre the multispecies intimacies, memories, and connections that were borne of, or that survived despite, the rise and wake of monoculture landscapes. Exemplifying this is Cuban artist Carlos Martiel's performance, *Sabiduría* (1989, *Wisdom*), that invokes and celebrates African-descendant spirituality through bodily mediated and empowering ties with the earth. Performed at Cuba's Academia de Artes Plásticas San Alejandro, *Sabiduría* features Martiel, dressed in white, kneeling on the lawn and engraving the Nyansapo symbol of West African wisdom into the grass using his tongue. The work reframes the earth as a medium of healing and ancestral communion, at the same time as it honours the resilience of enslaved peoples who trans-formed materials such as mud, clay, plants, bones, and ash into artistic and spiritual creativity and connection. In a similar vein, Malaysian artist Shooshie Sulaiman's *Tadika Getah* (2019, *Rubber Kindergarten*) and *Main Getah* (2019, *Rubberscape*), displayed at the Museum of Modern and Contemporary Art in Nusantara in Jakarta, draw on her cherished memories of growing up on a rubber plantation to challenge representations of the Malaysian rubber landscape solely as a colonial and capitalist space (Aroisoaie 2025). Instead, Sulaiman's art installations reclaim rubber as tradition, home, playground, kin, and education, creatively teasing out the ambivalent legacy of rubber in Malaysia as simultaneously charged with painful tensions and productive possibilities. Both pieces gesture towards the process of becoming-with rubber through an aesthetics of immersion that opens creative pathways for decolonial ways of living in and with plantations as sites of unexpected multispecies relatedness. The theme of plantationocenic immersion explored in Sulaiman's art finds resonance in Singapore-born artist Simryn Gill's photograph series, *Vegetation* (1999), in which Gill masks and disguises herself as a range of plants to naturalize human figures into the landscape. Shots of Gill in a monocrop oil palm plantation, holding a huge oil palm fruit bunch, her head composed of blurry oil palm fronds, conjure the vengeful alien plants of Cold War sci-fi imaginaries. At the same time, they symbolize an epoch wherein palm oil's ubiquity in foods and commodities worldwide is effectively bringing all of us together in a global process of 'becoming palm' (Gill & Taussig 2017).

p. 54

4.5 Conclusions

The plantation-spurred material transformations, socio-ecological relations, and epistemic cross-pollinations examined in this section offer valuable empirical and conceptual insights into the contexts and contours of emergence as 'the appearance of unprecedented relational properties, which are not reducible to any of the more basic component parts that give rise to them' (Kohn 2015: 166). Across the realms of illness and healing, vegetal ontologies, identities and imaginaries, and art and aesthetics, new lifeworlds have emerged at the sites where former ones have been displaced or disfigured, engendering new modes of communing with matter, species, and ecosystems, as well as new ways of sustaining life in the long shadow of plantation regimes. At the same time, it is important to remind ourselves that emergence has not always nor necessarily been emancipatory and anti-speciesist in its thrust and pull. Just as plantations have given rise to counter-hegemonic understandings and practices of more-than-human future-building, so too have they been the font of anthropocentric, racializing, and instrumentalizing processes anchored in the exploitation of bodies and beings in the name of science, medicine, and progress.

p. 54–55

To think with emergences across historical, cultural, and epistemic contexts helps draw attention to the different ethico-moral valences and eco-social consequences of knowledge production across the uneven power fields within which it takes place, takes hold, and meets resistance. It points to the continuously shifting ontological and representational grounds of plantations as multispecies contact zones where the emergence of new political and cultural figurations is tethered to the fates and futures, real and speculative, of both human and non-human actors. It also uncovers in agro-industrial landscapes unexpected possibilities for acts of collective and creative defiance that range from sweeping social movements to everyday acts of little justice. Taken together, these forms of defiance offer vital avenues for reckoning with the uneven conditions of more-than-human liveability that shape plantation lifeworlds and afterlives.

p. 55

5 Conclusion

The multidisciplinary field of the environmental humanities, to which this Element contributes, is premised on a rejection of the fictive yet powerful binarization of nature and culture. It is driven by the growing imperative to bring questions of meaning, value, ethics, politics, and justice to bear on environmental issues, at a time when ecological crises are threatening the lives and futures of humans and non-humans at a planetary scale (Rose et al. 2012). In engaging with knowledges across the sciences, arts, and humanities, the environmental humanities foreground how other-than-human agencies participate in the making and unmaking of multispecies worlds. In doing so, the current pushes against the (equally fictive and powerful) construct of human exceptionalism while also seeking a more nuanced understanding of the differential constructs of the 'human' across sites, scales, and subjects.

p. 56

It is in this spirit that the Element before you has offered to its readers an account of plantations that challenges the strict separation of humans as subjects from non-humans as objects – a separation that is itself fundamental to plantation logic –and to reframe them instead as mutually intertwined and consequential agents. Extraction, extinction, and emergence have acted as our material and conceptual entry points into the often-vexed multispecies landscapes of plantations past and present, drawing our attention to the imbrication of human and other-than-human lives and deaths in a range of different contexts – from territorial displacement and dispossession to enslavement and exploitation, resistance and failure, biodiversity loss and conservation, plantation pathologies and systems of healing, identity-formation and social movements, and artistic expression as political act. In this Conclusion, I reflect on how the materials and analyses presented in this Element invite attention to plantations through their onto-epistemic multiplicity, or what I term the 'plantation multiple'. I consider how such a framing might support the crafting of narratives that engage with plantations through their inter- and intra-species entanglements, that situate these entanglements against realities of plantationocenic violence beyond agricultural fields, and that unearth in plantations unexpected yet urgent possibilities for cultivating counter-plantation futures and multispecies justice.

5.1 The Plantation Multiple

Readers will have detected recurring motifs across the mosaic of plantation lifeworlds and afterlives examined in this Element. For instance, plantations globally have tended to emerge in close connection to broader national and transnational histories of capital, colonialism, and conquest (Araya 2023). Plantations constitute agricultural, economic, and labour institutions as much as they are political, cultural, and racializing structures, premised on human and vegetal settlement that has been instrumental to European expansionism and the spread of the capitalist world system (Thompson 1975). High levels of capital investment, large-scale market orientations, and a focus on land accumulation and concentration are central to plantation topographies and economies (Wolf & Mintz 1957). Plantations are characterized by a productionist logic anchored in the homogenization of landscapes, the exploitation of human and non-human labour, and the displacement of lifeforms and lifeways that obstruct imperial visions of progress, modernity, and mastery over natures-turned-profit. Just as important to the story of plantations is the resistance of places and peoples subject to the extinctions and extractions at the heart of plantation systems, and which I analysed in this Element through the lens of emergence.



Figure 5 A mature cotton field at harvest time, with open white bolls. Image credit: Mark Stebnicki.

Yet, despite their apparent homogeneity and hegemony, plantations are also as diverse in their composition and dynamics as the spatialities, societies, and temporalities upon and within which they have gained ground and grown (see Figure 5). In other words, there is no such thing as 'the' plantation in an abstract or generic sense (Clukey & Wells 2016; Chao 2022c). Rather, plantations exist in the plural. They are situated infrastructures that come into differential being and mattering through their variously recursive and refractive enactments in place and in time, or what one might term a *plantation multiple*. p. 57

The multiplicity at the heart of plantations ⁻one that belies their veneer of universality ⁻arises from factors spanning different praxiologies, ontologies, and epistemologies. These include the scales and styles of farming involved, the identities and affordances of cash crops cultivated, the ecosystems within which plantations flourish or flail, the communities and institutions implicated in their expansion, and the local and global ideologies and imaginaries alternately driving or impeding their proliferation. The space and time-specific contexts and contours of plantations also encompass the situated ways in which they have been apprehended at the grassroots and the situated knowledge systems that have come into generative friction or mutual accommodation as natures transform into resources ⁻the former ranging from outright or subversive opposition to desired or negotiated incorporation (Li 2007; Dove 2011; Hall et al. 2015; Mier & Cacho 2016), and the latter from scientific, agronomic, and technocratic epistemologies to cultural, spiritual, and moral ways of knowing, being, and relating (Mathews 2009; Whitaker 2020; Paredes 2021).

As participants of the 'plantation multiple', human and non-human actors co-shape and co-inhabit these planes of enactment through their different roles and relations. The perspectives and positionalities of these various human and other-than-human entities come to shape what counts as extraction, extinction, or emergence in the first instance, how they are experienced, and how they are responded to. To attend to the situatedness of these processes only further multiplies their affective, ethical, material, and historical valences, informing our understanding of plantations as multispecies contact zones, replete with more-than-human violence and vulnerability, and resistance and resilience. p. 58

5.2 Inter- and Intra-Species Entanglements

As a conceptual analytic, the 'plantation multiple' raises challenging but vital questions around how to do justice within environmental humanities narratives to the conjoined inter- and intra-species dimensions of monocrop systems and societies. These questions are prefigured in critiques of the Plantationocene stemming from critical race scholars, who call out the elision within recent plantation literature of long-standing and diverse bodies of work produced by Black intellectuals and activists on plantations and their bodily and ecological effects and affects (Weheliye 2014; Davis et al. 2019). They also amplify broader criticisms of multispecies approaches that, in seeking to centre non-human actors, run the risk not only of further invisibilising those racialized and exploited humans who continue to be deemed sub-human or non-human by dominant institutions (Jackson 2015; Galvin 2018) but also of neglecting Indigenous philosophies, practices, and protocols that have long been anchored in a recognition of the relational co-becoming of humans and non-humans (TallBear 2015; Winter 2023). They further trouble the epistemic and ethical valences of comparative research and associated representational methods when it comes to understanding and storying plantation dynamics such as violence and enslavement, that are at once yet differently racist and speciesist, replete with material and symbolic affinities yet also fundamentally distinct and arguably incommensurable.

One avenue for addressing these and other 'dreaded comparisons' (Spiegel 1996) is offered to us by critical race and literary scholars who uncover in plantation-driven extraction, extinction, and emergence, radical possibilities for rethinking the category of the 'human' and 'non-human' in the first instance ⁻and from there, for reimagining possibilities for alliance, empathy, and solidarity that might arise from a recognition that inter- and intra-species violence *both* ultimately find root in the dominant logics of colonialism and capitalism (Jackson 2020). This approach enables us to consider how plantation violence can be harnessed in the forging of new multispecies kinships, commonalities, potentialities, and promises, shared by humans deprived of personal history and non-humans stripped of their capacity to exist as autonomous historical actors (Bennett 2020). It challenges a sequential approach to race and nature (or vice versa) through an intersectional approach that approaches raciality and animality (alongside vegetality) as companion concepts and mutually addressable realities, from which new forms of multispecies defiance and resistance can arise (Boisseron 2018). It also allows us to delve into the contiguous and intersecting histories that have shaped the animalization of certain humans while also sustaining what Emily Yates-Doerr (2019: 297) calls a practice of 'careful equivocation'. Such a practice does not presume or seek sameness or unity across lifeforms and lifeways, nor results in the flattening of difference. Rather, it pays critical attention to the irreducibility of both race and species, in a world where what and who counts as 'human' remains deeply contested. p. 58–59

It is perhaps here that ecography, as a representational mode and narrative genre, might help enrich and nuance the stories we tell about plantation lifeworlds and afterlives. As my colleagues and I describe in a recent conversation on the topic (Bubandt et al. 2026), ecography underlines a shift in methodological focus from people (the ethnos in ethnography) to *systems of relations*, which encompass both human and other-than-human beings. By considering more-than-human worlds as homes (the ancient Greek root of eco-, 'oikos') for multiple different lifeforms, an eco- p. 59

graphic approach invites us to methodologically explore hierarchies and structures of relatedness within multispecies worlds and their deep connections to the economy (a term itself derived from 'oikos'), including its constitutive webs of cohabitation and care, and its patterns of extraction, extinction, and emergence. This mode of storytelling calls for critical and thick descriptions of more-than-human social worlds (Tsing 2014). It demands an attentiveness to situated, multispecies co-becomings that recognizes non-humans' agency without requiring their intentionality (van Dooren, Kirksey, & Münster 2016). It calls on us to acknowledge non-humans'subjectivities without presuming access to their experiential lifeworlds (Neimanis 2023). And it invites us to engage with the varying scales of violence and vulnerability engendered by plantations without presuming neat scalability across geographies, temporalities, and communities (Tsing 2012; Hecht 2018).

p. 59–60 Collectively reckoning with, repairing, and reimagining plantation and counter-plantation lifeworlds and afterlives, then, requires multispecies and relational stories, transdisciplinary cross-pollinations, and emplaced and reflexive analyses (Chao & Enari 2021). These approaches demand nothing less than an embrace of decolonial research methodologies that productively and collaboratively move beyond what Mythri Jegathesan (2021) terms 'disciplinary closets' by engaging meaningfully with the broader scholarly landscapes that plantation narratives are embedded in and informed by. In the process, we may uncover radical possibilities not only for the making of less evenly shared futures, but also for the decolonization of the broader climate crisis and attendant climate colonialities within which plantation pasts and presents are inextricably embedded (Sultana 2024).

5.3 (Counter)plantation Futures and Multispecies Justice

p. 60 While this Element has taken as its object of inquiry the processes of multispecies extraction, extinction, and emergence shaping plantation systems and societies, it is important to recall that plantation logic extends well beyond the realm of literal agricultural fields. Embodying what Billy C. Hawkins (2010) calls 'the new plantation', this logic infiltrates in often unwitting or under-acknowledged ways, a range of contemporary social, political, and economic structures. In the United States, Jamaica, Brazil, Barbados, and other world geographies, the afterlives of colonial plantation capitalism manifest in the concatenation of White-dominated elite institutions, including the university (Williams 2021); discriminatory carceral institutions, such as the prison (McKittrick 2011, 2013); racializing legal structures, including the criminal system (DeLombard 2019); unequal urban infrastructures and violence, including slums and state-surveillance (McKinson 2024); exploitative and history-effacing entertainment industries, including athletics and tourism (Hawkins 2010; Perry 2013); as well as everyday experiences of metabolic, psychological, and racial violence among descendants of enslaved or indentured plantation workers (Hartman 1997, 2007; Bahadur 2014; Hatch, Sternlieb & Gordon 2019; Moran-Thomas 2019).

p. 60–61 Just as people and ecosystems have long challenged the homogenizing, racializing, and disciplining force of plantation logic in the making of what Jean Casimir (2020) calls 'counterplantation systems', so too are many struggles for justice today being led by grassroots communities living in the wake of the plantation – whether in the form of anti-racism advocacy, prison and police abolitionist movements, or local food sovereignty initiatives that defy the hegemony of corporate agro-industrial production systems (Dubois & Turits 2019; Williams, Squire & Tuitt 2021). Both in the context of monocrop lifeworlds and their afterlives, then, it is critical to acknowledge that the unearthing of plantation alternatives – or counterplantation futures – stems not only from scholarly spaces, but rather sits on the shoulders of thousands of activists across time and around the world who have fought – and who continue to fight – for access to land and freedom, including the right to inhabit, own, manage, cultivate, work, and dream about such lands according to the principles of sovereignty and self-determination (Chao et al. 2024: 546). Again, one can identify the influence of such counterplantation imaginaries in spaces and contexts well beyond monocrop fields themselves – from the civil rights movement and subsequent recognition of constitutional rights in the United States which took as one of its root aims the abolishment of 'plantation mentality' and its race, gender, and power dynamics (Green 2007), to the very idea and institutionalization of universal rights, which is often understood as a Global North/European concept, but that was in fact driven first and foremost by the struggles by people of colour over enslavement and citizenship in the French Caribbean (Dubois 2012).

p. 61–62 The rippling ramifications and manifestations of the 'plantation multiple' bring us to reflect on whether there is, or has ever been, any agro-capitalist form that is not a plantation in one form or another, and how thinking about these forms collectively yet differently *as* plantations might uncover novel insights into their impacts on, and implications for, human and other-than-human lifeforms. These ramifications and manifestations draw attention to a a potentially productive conceptual tension in works such as the one before you, which seek to analyse spatially, temporally, and materially disparate plantation forms through over-arching conceptual frameworks (in this instance, extraction, extinction, and emergence) and, in doing so, risk flattening plantations' proclaimed multiplicity through their very own classificatory schemes – even as these schemes are not mutually exclusive, exhaustive, or fixed. In real-world terms, the plantation multiple fosters an understanding and realization that while we ourselves may not inhabit or be directly affected by industrial monocrop landscapes, all of us, in one way or another, still inhabit the shadow of the plantation

through the institutions, ideologies, imaginaries, and materialities that surround us and shape us –from the ingredients in the foods that we consume, to the materials of the clothes that we wear, to the biofuels that we choose as ethical substitutes for non-renewable energy. Given the far-ranging and yet often invisibilized presence of plantations and plantation commodities in the contemporary world, one might ask: what modes of everyday and institutional repair are required in imagining and enacting futures that are not predicated on human exploitation and environmental exhaustion as the preconditions for endless capitalist growth (Thomas 2019; Cousins 2023)? What social and ecological reconfigurations or revolutions are required for agriculture to enable, rather than efface, the meaningful connections between people, plants, and landscapes that nourish human and other-than-human life (Flachs 2025; Pascoe 2014)? What lessons can counterplantation future-making projects across the world teach us about living, thinking, remembering, storying, hoping, and relating beyond the paralysing frame of apocalyptic narratives in this age of radical ecological unmaking (McKittrick 2013; Moulton & Harris 2025)? How might asking and addressing these questions collectively help us challenge the normative violence of the Plantationocene and together inhabit instead what Natasha Myers (2017: 300) terms the 'Planthroposcene', or a way of living and thinking that recognizes humans' profound interimplications with the vegetal world and that is attuned to plants' own desires and demands outside the rhythms of capitalist extraction?

The questions above are, fundamentally, *questions of justice* –justice for differently situated human communities, but also justice for differently situated non-human organisms, collectives, environments, and landscapes, or what has, in recent years, been theorized as 'multispecies justice' (Celermajer et al. 2020; Tschakert et al. 2021; Chao, Bolender & Kirksey 2022; Celermajer et al. 2025). As a transformational rather than extensionist project, multispecies justice invites us to reframe the subject of justice beyond the human to encompass the multiplicity of plants, animals, and ecosystems whom humans become with, while also attending to enduring and emergent social, racial, gendered, economic, and climate injustices. It is a project that recognizes that justice in more-than-human worlds is always patchy and partial –more for some beings than for others (Chao & Kirksey 2022). It is a project that demands we ask who benefits from current configurations and distributions of justice, what these configurations reveal about which lives count and what counts as life, and how best to address the profound imbrications of non-human life with human politics, culture, and institutions (Chao & Celermajer 2023). And it is a project that acknowledges that no single form of writing, scholarly field, or perspective can adequately or comprehensively contain or convey the intersectional entanglements, possible alliances, and ethical stakes of justice-making projects in more-than-human worlds (Bolender, Chao & Kirksey 2022: 230). p. 62

In the context of the 'plantation multiple' and its complex weave of extraction, extinction, and emergence, the pursuit of multispecies justice as principle and practice is also one that demands that we take the specificity and situatedness of each plantation lifeworld and afterlife as a starting point in reimagining both the life of matter and the matter of life (Chao, Winter & Schlosberg 2026). Holding on to this specificity and situatedness challenges the vision and violence of plantation logics as a simplifying and scale-making project, anchored in the fiction of human control over putatively passive landscapes turned Cheap Natures (Moore 2015: 234). Instead it paves the way towards a different kind of multispecies future-building –one that draws on the experiences, lessons, and struggles of people around the world who are most directly and deeply mired by the predicament of Plantationocenic living and dying and its conjoined dynamics of extraction, extinction, and emergence. p. 62–63

It is perhaps in such a pursuit for multispecies justice that we might uncover glimpses of hopeful promise and possibility for the kind of universal envisioned by Martinican author Aimé Césaire (1955: 6) in his critique of White European privilege. This universal is not, as plantation logic insists, premised on the pursuit of simplification to the ends of efficiency and control. Rather, it is a universal that radically defies plantation logic by embracing differences of all kinds and the difference these differences make –a universal, in Césaire's own words, that is 'rich with all that is particular, rich with all the particulars there are, the deepening of each particular, the coexistence of them all'. p. 63

References

p. 64 Aaron-Denz, M. (2021). ‘Health Knowledge of Former Days in a New Era’. In H. Menke, J. Buckingham, F. Gounder, A. Kumar, & M. S. Hassankhan (eds.), *Social Aspects of Health, Medicine and Disease in the Colonial and Post-Colonial Era* (157–178). Routledge.

Aga, A. (2022). *Genetically Modified Democracy: Transgenic Crops in Contemporary India*. Yale University Press.

Agard-Jones, V. (2013). ‘Bodies in the System’. *Small Axe*, *42*, 182–192.

Aikens, N., Clukey, A., King, A. K. & Wagner, I. (2019). ‘South to The Plantationocene’, *ASAP/Journal*. http://asapjournal.com/south-to-the-plantationocene-natalie-aikens-amy-clukey-amy-k-king-and-isadora-wagner/.

Allewaert, M. (2013). *Ariel’s Ecology: Plantations, Personhood, and Colonialism in the American Tropics*. University of Minnesota Press.

Araya, A. L. (2023). ‘Monocrops’. *The Journal of Peasant Studies*, *50*(3), 797–808.

Arosoaie, A. (2025). ‘Aesthetics of Immersion: Contemporary Art and Pedagogies of Life in the Malaysian Rubberscape’. *Environmental Humanities*, *17*(2), 313–334.

Aso, M. (2018). *Rubber and the Making of Vietnam: An Ecological History, 1897–1975*. University of North Carolina Press.

Bahadur, G. (2014). *Coolie Woman: The Odyssey of Indenture*. University of Chicago Press.

Baptista, J. B. (2018). ‘Eco(il)logical Knowledge: On Different Ways of Relating with the Known’. *Environmental Humanities*, *10*(2), 397–420.

Barua, M. (2004). *Plantation Worlds*. Duke University Press.

Barua, M. (2023). ‘Plantationocene: A Vegetal Geography’. *Annals of the American Association of Geographers*, *113*(1), 13–29.

Barua, M. (2024). *Plantation Lifeworlds*. Duke University Press.

Bastos, C. (2024). ‘Plant-People Intimacies: Sugar Canes, Pineapples and the Memory of Migration in Hawai‘i’. *Journal of Ethnobiology*, *44*(1), 11–22.

Beilin, K. O. & Suryanarayanan, S. (2017). ‘The War between Amaranth and Soy: Interspecies Resistance to Transgenic Soy Agriculture in Argentina’. *Environmental Humanities*, *9*(2), 204–229.

Benítez-Rojo, A. (1996). *The Repeating Island: The Caribbean and the Postmodern Perspective*. Translated by J. E. Maraniss. Duke University Press.

p. 65 Bennett, J. (2020). *Being Property Once Myself: Blackness and the End of Man*. Harvard University Press.

Besky, S. (2013). *The Darjeeling Distinction: Labor and Justice on Fair-Trade Tea Plantations in India*. University of California Press.

Besky, S. (2020). ‘Monoculture.’ in C. Howe & A. Pandian (eds.) *Anthropocene Unseen: A Lexicon* (277–280). Punctum Books.

Bindas, K. (1992). *America’s Musical Pulse: Popular Music in Twentieth-Century Society*. Bloomsbury.

Boisseron, B. (2018). *Afro-Dog: Blackness and the Animal Question*. Columbia University Press.

Bolender, K., Chao, S., & Kirksey, E. (2022). ‘Afterword: Fugitive Jurisdictions’. In S. Chao, K. Bolender, & E. Kirksey (eds.) *The Promise of Multispecies Justice* (229–238). Duke University Press.

Borras Jr, S. M., & Franco, J. C. (2011). ‘Global Land Grabbing and Trajectories of Agrarian Change: A Preliminary Analysis’. *Journal of Agrarian Change*, *12*(1), 34–59.

Bourgois, P. (1989). *Ethnicity at Work: Divided Labor on a Central American Banana Plantation*. Johns Hopkins University Press.

Brandão, J., Cardoso, F. & Garrett, R. (2025). ‘Why Has the Brazilian Cerrado Been Left Behind by Voluntary Environmental Policies?’ *Global Environmental Change*, *92*, 103005.

Brockington, D. (2002). *Fortress Conservation: The Preservation of the Mkomazi Game Reserve, Tanzania*. James Currey Publishers.

Bubandt, N., Sophie, C. Marianne, L. et al. (2026). ‘Anthropologists Are Talking about Ecography’. *Ethnos*, *91*(1), 126–143.

Butler, J. (2004). *Precarious Life: The Powers of Mourning and Violence*. Verso.

Byerlee, D., Stevenson, J., & Villoria, N. (2014). 'Does Intensification Slow Crop Land Expansion or Encourage Deforestation?' *Global Food Security*, *3* (2), 92–98.

Cardoso, T. (2016). 'The Multispecies Life of Feral Dendezeiros: Ethnography in Motion'. *Engagement Blog*. https://aesengagement.wordpress.com/2016/02/02/the-multispecies-life-of-feral-dendezeiros-ethnography-in-motion/.

Carlson, K. M., & Curran, L. M. (2013). 'Refined Carbon Accounting for Oil Palm Agriculture: Disentangling Potential Contributions of Indirect Emissions and Smallholder Farmers'. *Carbon Management*, *4*(4), 347–349.

Casimir, J. (2020). *The Haitians: A Decolonial History*. Translated by L. Dubois. University of North Carolina Press.

Celermajer, C., Burke, A. Fishel, S. et al. (2025). *Institutionalizing Multispecies Justice*. Cambridge University Press.

p. 66

Celermajer, D. Schlosberg, D., Rickards, L. et al. (2020). 'Multispecies Justice: Theories, Challenges, and a Research Agenda for Environmental Politics'. *Environmental Politics*, *30*(1–2), 119–140.

Césaire, A. (1955). *Discourse on Colonialism*. Monthly Review Press.

Césaire, A. (2001). *Notebook of a Return to the Native Land*. Translated by C. Eshleman & A. Smith. Wesleyan University Press.

Chao, S. (2019). The Truth About 'Sustainable' Palm Oil. *SAPIENS*. June 13. www.sapiens.org/culture/palm-oil-sustainable/.

Chao, S. (2020). 'ATree of Many Lives: Vegetal Teleontologies in West Papua'. *HAU: Journal of Ethnographic Theory*, *10*(2), 514–529.

Chao, S. (2021a). 'The Beetle or the Bug? Multispecies Politics in a West Papuan Oil Palm Plantation'. *American Anthropologist*, *123*(2), 476–489.

Chao, S. (2021b). 'Can There Be Justice Here? Indigenous Perspectives from the West Papuan Oil Palm Frontier'. *Borderlands*, *20*(1), 11–48.

Chao, S. (2021c). 'Children of the Palms: Growing Plants and Growing People in a Papuan Plantationocene'. *Journal of the Royal Anthropological Institute*, *27*(2), 245–264.

Chao, S. (2022a). *In the Shadow of the Palms: More-Than-Human Becomings in West Papua*. Duke University Press.

Chao, S. (2023b). 'Multispecies Mourning: Grieving as Resistance on the West Papuan Oil Palm Frontier'. *Cultural Studies*, *37*(4), 553–579.

Chao, S. (2022c). 'Plantation'. *Environmental Humanities*, *14*(2), 361–366.

Chao, S. (2022d). '(Un)Worlding the Plantationocene: Extraction, Extinction, Emergence'. *ETropic: Electronic Journal of Studies in the Tropics*, *21*(1), 165–191.

Chao, S. (2023). 'Spent Earth'. *Critical Times*, *6*(2), 179–188.

Chao, S. (2024). 'Spaces and Subjects of Rupture on the Papuan Oil Palm Frontier.' *Antipode Online*. 4 January. https://antipodeonline.org/wp-content/uploads/2023/12/1_Chao.pdf.

Chao, S. (2025). *Land of Famished Beings: West Papuan Theories of Hunger*. Duke University Press.

Chao, S. (2026). 'Thinking Beyond Bios in an Age of Planetary Unmaking'. In S. Chao, C. Winter & D. Schlosberg (eds.), *Worlds Beyond Bios: The Life of Matter and the Matter of Life* (1–26). Duke University Press.

Chao, S., Wolford, W., Ofstehage, A., Guttal, S., Gonçalves, E., & Ayala, F. (2024). 'The Plantationocene as Analytical Concept: A Forum for Dialogue and Reflection.' *The Journal of Peasant Studies*, *51*(3), 541–563.

Chao, S., & Celermajer, D. (2023). 'Introduction: Multispecies Justice'. *Cultural Politics*, *19*(1), 1–17.

p. 67

Chao, S., & Kirksey, E. (2022). 'Introduction: Who Benefits from Multispecies Justice?' In S. Chao, E. S. Kirksey, & K. Bolender (eds.), *The Promise of Multispecies Justice* (1–22). Duke University Press.

Chao, S., & Enari, D. (2021). 'Decolonising Climate Change: A Call for Beyond-Human Imaginaries and Knowledge Generation'. *ETropic: Electronic Journal of Studies in the Tropics*, *20*(2), 32–54.

Chao, S., Krupa, C., & Li, T. M. (2024). 'Anthropologists are Talking –About Contemporary Plantations. Technologies, Violence, and Vulnerability Across Geographies and Genealogies'. *Ethnos*, *89*(1), 158–187.

Chao, S., Bolender, K., & Kirksey, E. (2022). *The Promise of Multispecies Justice*. Duke University Press.

Chamoiseau, P. (1994). *Creole Folktales*. Translated by L. Coverdale. New Press.

Child, B. (2019). 'The Plantation Countermelodies of Dunbar and Du Bois: Writing Agropolitical Subjecthood in the Nadir.' *American Literature*, *91*(3), 557–586.

Chua, L., Fair, H., Schreer, V., Stepien, A., & Thung, H. (2021). '"Only the Orangutans Get a Life Jacket": Uncommoning Responsibility in a Global Conservation Nexus'. *American Ethnologist*, *48*(4), 370–385.

Clukey, A., & Wells, J. (2016). 'Introduction: Plantation Modernity'. *The Global South*, *10*(2), 1–10.

Colchester, M., Chao, S. & Jiwan, N. (2012). *Securing Rights through Commodity Roundtables? A Comparative Review*. Forest Peoples Programme. https://www.forestpeoples.org/fileadmin/uploads/fpp/migration/publication/2012/11/securing-rights-through-commodity-roundtables-comparative-review.pdf.

Colchester, M., S. Chao, & P. Anderson (2014). *Assault on the Commons: Deforestation and Denial of Rights in Indonesia*. Moreton-in-Marsh: Forest Peoples Programme. https://www.forestpeoples.org/fileadmin/uploads/fpp/migration/publication/2014/12/assault-commons.pdf.

Conde, M. (2020). 'A Philippine Stingless Bee Helps Boost Coconut Yields and Empower Women'. *Mongabay*, 29 September. https://news.mongabay.com/2020/09/a-philippine-stingless-bee-helps-boost-coconut-yields-and-empower-women/.

Conway, G. (2012). *One Billion Hungry: Can We Feed the World?* Cornell University Press.

Cousins, T. (2023). *The Work of Repair: Capacity After Colonialism in the Timber Plantations of South Africa*. Fordham University Press.

Covington, J. W. (2001). *The Seminoles of Florida*. University Press of Florida.

p. 68 Cramb, R. & McCarthy, J. (2016). 'Characterising Oil Palm Production in Indonesia and Malaysia'. In R. Cramb & J. McCarthy (eds.), *The Oil Palm Complex: Smallholders, Agribusiness and the State in Indonesia and Malaysia* (27–77). National University of Singapore Press.

Cronon, W. (1996). 'The Trouble with Wilderness: Or, Getting Back to the Wrong Nature'. *Environmental History*, *1*(1), 7–28.

Curry, G. N., & Koczberski, G. (2009). 'Finding Common Ground: Relational Concepts of Land Tenure and Economy in the Oil Palm Frontier of Papua New Guinea'. *The Geographical Journal*, *175*(2), 98–111.

Danielsen, F., Beukema, H., Burgess, N. D. et al. (2009). 'Biofuel Plantations on Forested Lands: Double Jeopardy for Biodiversity and Climate'. *Conservation Biology*, *23*, 348–358.

Dauvergne, P., & Neville, K. J. (2010). 'Forests, Food, and Fuel in the Tropics: The Uneven Social and Ecological Consequences of the Emerging Political Economy of Biofuels'. *The Journal of Peasant Studies*, *37*(4), 631–660.

Davis, J., Moulton, A. A., van Sant, L., & Williams, B. (2019). 'Anthropocene, Capitalocene, . . . Plantationocene?: A Manifesto for Ecological Justice in an Age of Global Crises'. *Geography Compass*, *13*(e12438), 1–15.

Day, L., & Hogue, R. (2019). "Plantation Housing Isn't the Answer to Homelessness in Hawai'i." *Edge Effects*. http://edgeeffects.net/kahauiki-village/.

de la Cadena, M. (2014). 'Runa: Human But Not Only'. *HAU: Journal of Ethnographic Theory*, *4*(2), 253–259.

Dean, W. (1987). *Brazil and the Struggle for Rubber: A Study in Environmental History*. Cambridge University Press.

DeLombard, J. (2019). 'Dehumanizing Slave Personhood'. *American Literature*, *19*(3), 491–521.

Dillon, E. M. (2019). 'The Plantationocene and the Performative Commons: A Brief History of Uncommoning'. *Minnesota Review*, *93*, 83–93.

Dove, M. R. (2011). *The Banana Tree at the Gate: A History of Marginal Peoples and Global Markets in Borneo*. Yale University Press.

Driessen, (2010). *Eco-Imperialism: Green Power, Black Death*. Merril Press.

Dubois, L. (2012). *A Colony of Citizens: Revolution and Slave Emancipation in the French Caribbean, 1787–1804*. University of North Carolina Press.

Dubois, L., & Turits, R. (2019). *Freedom Roots: Histories from the Caribbean*. University of North Carolina Press.

Dunbar, P. L. (1903). 'The Haunted Oak'. In J. M. Braxton (ed.), *The Collected Poetry* (219–220). University of Virginia Press.

p. 68–69 Eersel, M. (2021). 'Revisiting F.A. Kuhn's "Reflection on the Situation of the Surinamese Plantation Slaves: An

Economic-Medical Contribution to its Improvement (1828)"'. In Menke, H., J. Buckingham, F. Gounder, A. Kumar, & M. S. Hassankhan (eds.), *Social Aspects of Health, Medicine and Disease in the Colonial and Post-colonial Era* (75–100). Routledge.

Eichen, J. (2020). 'Cheapness and (Labor-)Power: The Role of Early Modern Brazilian Sugar Plantations in the Racializing Capitalocene'. *Environment and Planning D: Society and Space*, *38*(1), 35–52.

p. 69

Escobar, A. (1999). 'After Nature: Steps to an Antiessentialist Political Ecology'. *Current Anthropology*, *40*(1), 1–30.

Fabro, K. A. (2023). 'Is the Genetically Modified, Nutrient-rich Golden Rice as Safe as Promised?' *Mongabay*. 11 September. https://news.mongabay.com/2023/09/is-the-genetically-modified-nutrient-rich-golden-rice-as-safe-as-promised/.

Fett, S. (2002). *Working Cures: Healing, Health, and Power on Southern Slave Plantations*. University of North Carolina Press.

Fitzherbert, E. B., Struebig, M. J., Morel, A. et al. (2008). 'How Will Oil Palm Expansion Affect Biodiversity?' *Trends in Ecology and Evolution*, *23*(10), 539–545.

Flachs, A. (2025). *Feeding the World as If People Mattered: How Small Farms Produce Value Beyond Yields*. University of Arizona Press.

Flachs, A., Bastos, C., Heath, D., & Venkateswar, S. (2024). 'Introduction to Special Collection: Plant-Anthropo-Genesis: The Co-Production of Plant–People Lifeworlds.' *Journal of Ethnobiology*, *44*(1), 3–10.

Forney, J., Rosin, C., & Campbell, H. (2018). *Agri-Environmental Governance as an Assemblage: Multiplicity, Power, and Transformation*. Routledge.

Foster, L.A. (2017). *Reinventing Hoodia: Peoples, Plants, and Patents in South Africa*. University of Washington Press.

Gabriel, J., Nelson, N., Filer, C., & Wood, M. (2017). Oil Palm Development and Large-Scale Land Acquisitions in Papua New Guinea. In S. McDonnell, M. G. Allen, & C. Filer (eds.), *Kastom, Property, and Ideology: Land Transformations in Melanesia* (205–250). Australian National University Press.

Galvin, S. S. (2018). 'Interspecies Relations and Agrarian Worlds.' *Annual Review of Anthropology*, *47*, 233–249.

Giesen, J. (2011). *Boll Weevil Blues: Cotton, Myth, and Power in the American South*. University of Chicago Press.

Gill, S., & Taussig, M. (2017). *Becoming Palm*. NTU Centre for Contemporary Art Singapore & Sternberg Press.

Gilroy, P. (1995). *The Black Atlantic: Modernity and Double Consciousness*. Harvard University Press.

Goldstein, J. (2016). 'Carbon Bomb: Indonesia's Failed Mega Rice Project'. *Environment & Society Portal*, Arcadia, 6. https://doi.org/10.5282/rcc/7474.

p. 70

Gordillo, G. R. (2004). *Landscapes of Devils: Tensions of Place and Memory in the Argentinean Chaco*. Duke University Press.

Gordillo, G. R. (2014). *Rubble: The Afterlife of Destruction*. Duke University Press.

Grandin, G. (2010). *Fordlandia: The Rise and Fall of Henry Ford's Forgotten Jungle City*. Picador.

Gray, R., & Sheikh, S. (2018). 'The Wretched Earth: Botanical Conflicts and Artistic Interventions. Introduction'. *Third Text*, *32*(2–3), 163–175.

Green, L. (2007). *Battling the Plantation Mentality: Memphis and the Black Freedom Struggle*. University of North Carolina Press.

Guthman, J. (2019). *Wilted: Pathogens, Chemicals, and the Fragile Future of the Strawberry Industry*. University of California Press.

Hall, R., Edelman, M., Borras, S., White, B., & Wolford, W. (2015). 'Resistance, Acquiescence or Incorporation? An Introduction to Land Grabbing and Political Reactions "From Below"'. *The Journal of Peasant Studies*, *42*(3–4), 467–488.

Han, S., & McAndrew, J. (2018). 'Impact of Government Policies and Corporate Land Grabs on Indigenous People's Access to Common Lands and Livelihood Resilience in Northeast Cambodia'. *Land*, *7*(4), 122.

Haraway, D. J. (2015). 'Anthropocene, Capitalocene, Plantationocene, Chthulucene: Making Kin'. *Environmental Humanities*, *6*, 159–165.

Hartman, S. (1997). *Scenes of Subjection: Terror, Slavery, and Self-Making in Nineteenth Century America*. Oxford University Press.

Hartman, S. (2007). *Lose Your Mother: A Journey Along the Atlantic Slave Route*. Farrar, Strauss & Giroux.

Harvey, D. (2003). *The New Imperialism*. Oxford University Press.

Hatch, A. R., Sternlieb, S., & Gordon, J. (2019). 'Sugar Ecologies: Their Metabolic and Racial Effects'. *Food, Culture, and Society*, *22*(5), 595–607.

Hatch, T. (2012). *Osceola and the Great Seminole War*. St. Martin's Press.

Hetherington, K. (2013). "Beans Before the Law: Knowledge Practices, Responsibility, and the Paraguayan Soy Boom." *Cultural Anthropology*, *28* (1), 65–85.

Hawkins, B. (2010). *The New Plantation: Black Athletes, College Sports, and Predominantly White NCAA Institutions*. Palgrave Macmillan.

Hecht, G. (2018). 'Interscalar Vehicles for an African Anthropocene: On Waste, Temporality, and Violence'. *Cultural Anthropology*, *33*(1), 109–141.

p. 71 Herrmans, I. (2020). 'Spirits Out of Place: Relational Landscapes and Environmental Change in East Kalimantan, Indonesia'. *Journal of the Royal Anthropological Institute*, *26*(4), 766–785.

Hetherington, K. (2019). 'The Concentration of Killing: Soy, Labor, and the Long Green Revolution'. In S. Besky & A. Blanchette (eds.), *How Nature Works: Rethinking Labor on a Troubled Planet* (41 – 58). University of New Mexico Press.

Hetherington, K. (2020). *The Government of Beans: Regulating Life in the Age of Monocrops*. Duke University Press.

Hernandez, J. (2022). *Fresh Banana Leaves: Healing Indigenous Landscapes through Indigenous Science*. North Atlantic Books.

Hiddleston, J. (2025). '"Everything is in Everything": Tropiques, Césaire, and Ecological Thought'. *ISLE: Interdisciplinary Studies in Literature and Environment*, *32*(1), 133–115.

Hopkins, J. F. (1951). *A History of the Hemp Industry in Kentucky*. University Press of Kentucky.

Hosbey, J., & Roane, J. T. (2021). 'ATotally Different Form of Living: On the Legacies of Displacement and Marronage as Black Ecologies'. *Southern Cultures*, *27*(1), 68–73.

Ives, S. (2017). *Steeped in Heritage: The Racial Politics of South African Rooibos Tea*. Duke University Press.

Jackson, S. (2012). *Creole Indigeneity: Between Myth and Nation in the Caribbean*. University of Minnesota Press

Jackson, Z. I. (2015). 'Outer Worlds: The Persistence of Race in Movement "Beyond the Human"'. *GLQ: A Journal of Lesbian and Gay Studies*, *21*(2–3), 215–218.

Jackson, Z. I. (2020). *Becoming Human: Matter and Meaning in an Antiblack World*. New York University Press.

Jegathesan, M. (2021). 'Black Feminist Plots before the Plantationocene and Anthropology's "Regional Closets"'. *Feminist Anthropology*, 2: 78–93.

Judith, C. (2004). '"With Grains in her Hair': Rice History and Memory in Colonial Brazil'. *Slavery and Abolition*, *25*(1), 1–27.

Keeve, C.B. (2020). "Fugitive Seeds." *Edge Effects*. Available at: https://edgeeffects.net/fugitive-seeds/.

Keil, P. (2025). *The Presence of Elephants: Sharing Lives and Landscapes in Assam*. Routledge.

Kelley, R. D. G. (2002). *Freedom Dreams: The Black Radical Imagination*. Beacon Press.

p. 72 Kennedy, R. (2013). *Cotton and Conquest: How the Plantation System Acquired Texas*. University of Oklahoma Press.

Kenney-Lazar, M., & Ishikawa, N. (2019). 'Mega-Plantations in Southeast Asia'. *Environment and Society*, *10*(1), 63–82.

King, A. (2016). '"Just like Back Home—Only Different!": Plantation Exploitation in 1970s Women-in-Prison Movies Filmed in the Philippines'. *The Global South*, *10*(2), 48–69.

King, T. L. (2016). 'The Labor of (Re)reading Plantation Landscapes Fungible(ly)'. *Antipode*, *48*(4), 1022–1039.

King, T. L. (2019). *The Black Shoals: Offshore Formations of Black and Native Studies*. Duke University Press.

Koczberski, G. and Curry, G.N. (2004). 'Divided Communities and Contested Landscapes: Mobility, Development, and Shifting Identities in Migrant Destination Sites in Papua New Guinea.' *Asia Pacific Viewpoint*, 45(4), 357–371.

Kohn, E. (2015). *How Forests Think: Toward an Anthropology beyond the Human*. University of California Press.

Koshy, S. L. M. Cacho, J. A. Byrd, & B. J. Jefferson (eds.) (2022). *Colonial Racial Capitalism*. Duke University Press.

Kröger, M. (2022). *Extractivisms, Existences, and Extinctions: Monoculture Plantations and Amazon Deforestation*. Routledge.

Krupa, C. (2022). *A Feast of Flowers: Race, Labor, and Postcolonial Capitalism in Ecuador*. University of Pennsylvania Press.

Kumpf, D. (2021). 'Multispecies Monocultures: Organic Agriculture and Resistance on Indian Tea Plantations'. *Berliner Blätter*, *84*, 49–61.

Kumpf, D. (2023). 'Finding Deities in Monoculture Rows: Everyday Affects of Green Growth'. *Antipode Online*. 4 January. https://antipodeonline.org/wp-content/uploads/2023/12/4_Kumpf.pdf.

Lattas, A. (1993). 'Sorcery and Colonialism: Illness, Dreams, and Death as Political Languages in West New Britain'. *Man*, *28*(1), 51–77.

Leguizamón, A. (2020). *Seeds of Power: Environmental Injustice and Genetically Modified Soybeans in Argentina*. Duke University Press.

Lévi-Strauss, C. (1961). *Tristes Tropiques*. Translated by J. Russell. Criterion Books.

Lewis, D. C. (1996). *The Plantation Dream: Developing British New Guinea and Papua 1884–1942*. The Journal of Pacific History.

Li, T. M. (2007). *The Will to Improve: Governmentality, Development, and the Practice of Politics*. Duke University Press.

Lima, M., da Silva Junior, C. A. Pelissari, T. D. et al. (2020). 'Sugarcane: Brazilian Public Politics Threaten the Amazon and Pantanal Biomes'. *Perspectives in Ecology and Conservation*, *18*(3), 210–212.

p. 73

Loichot, V. (2013). *The Tropics Bite Back: Culinary Coups in Caribbean Literature*. University of Minnesota Press.

López, A. J. (2011). 'The Plantation as Archive: Images of "the South" in the Postcolonial World'. *Comparative Literature*, *63*(4), 402–422.

Macedo, M. (2024). 'Plantations Beyond Monocrops: Cannabis Ecologies From Colonial Angola to São Tomé'. *Journal of Ethnobiology*, *44*(1), 46–55.

Manjapra, K. (2018). 'Plantation Dispossessions: The Global Travel of Agricultural Racial Capitalism'. In S. Beckert and C. Desan (eds.), *American Capitalisms: New Histories* (361–386). Columbia University Press.

Mathews, A. S. (2009). 'Unlikely Alliances: Encounters between State Science, Nature Spirits, and Indigenous Industrial Forestry in Mexico, 1926–2008'. *Current Anthropology*, *50*(1), 75–101.

Mathews, A. S. (2018). 'Landscapes and Throughscapes in Italian Forest Worlds: Thinking Dramatically about the Anthropocene'. *Cultural Anthropology*, *33*(3), 386–414.

Mbembe, A. (2003). 'Necropolitics.' Translated by S. Corcoran. *Public Culture*, *15*(1), 11–40.

McKie, R. (2024). '"A Catastrophe": Greenpeace Blocks Planting of "Lifesaving" Golden Rice'. *The Guardian*. May 25. www.theguardian.com/environment/article/2024/may/25/greenpeace-blocks-planting-of-lifesaving-golden-rice-philippines.

McKinson, K. (2024). 'A House Pon the Hill: Refractions of Black Death and Desire through the Jamaican Plantation Great House'. *Antipode Online*. 4 January. https://antipodeonline.org/wp-content/uploads/2023/12/9_McKinson.pdf.

McKittrick, K. (2011). 'On Plantations, Prisons, and a Black Sense of Place.' *Social & Cultural Geography*, *12*(8), 947–963.

McKittrick, K. (2013). 'Plantation Futures'. *Small Axe*, *17*(3), 1–15.

McInnis, J. C. (2016) '"Behold the Land": W. E. B. Du Bois, Cotton Futures, and the Afterlife of the Plantation in the US South.' *The Global South*, *10*(2), 70–98.

Meijaard, E., S. A. Wich. M. Ancrenaz., & A. J. Marshall. (2011). 'Not by Science Alone: Why Orangutan Conservationists Must Think Outside the Box'. *Annals of the New York Academy of Sciences*, *1249*(1), 29–44.

p. 74 Meijaard, E., J. Sherman, M. Ancrenaz, S. A. Wich, T. Santika, & M. Voigt. (2018). 'Orangutan Populations are Certainly Not Increasing in the Wild'. *Current Biology*, *28*(21), R1241–R124.

Meijaard, E., Garcia-Ulloa, J., & Douglas, S. (2018). *Oil Palm and Biodiversity: A Situation Analysis by the IUCN Oil PALM Task Force*. IUCN Oil Palm Task Force.

Menke, J., Ramdas, S., Reyme, M., & Pieters, T. (2021). 'Leprosy, a Multidimensional Approach: Colonialism, Slavery, Indentured Labour and Animal Mythology in Suriname'. In H. Menke, J. Buckingham, F. Gounder, A. Kumar, & M. S. Hassankhan (eds.), *Social Aspects of Health, Medicine and Disease in the Colonial and Post-colonial Era* (223–246). Routledge.

Mier, M., & Cacho, T. G. (2016). 'Soybean Agri-Food Systems Dynamics and the Diversity of Farming Styles on the Agricultural Frontier in Mato Grosso, Brazil'. *The Journal of Peasant Studies*, *43*(2), 419–441.

Milne, S. (2022). *Corporate Nature: An Insider's Ethnography of Global Conservation*. University of Arizona Press.

Mintz, S. W. (1996). *Tasting Food, Tasting Freedom: Excursions into Eating, Culture, and the Past*. Beacon Press.

Missouri Sherman-Peter, A. (2022). *The Legacy of Slavery in the Caribbean and the Journey Towards Justice*, *UN Chronicle*. www.un.org/en/un-chronicle/legacy-slavery-caribbean-and-journey-towards-justice.

Mitman, G. (2021). *Empire of Rubber: Firestone's Scramble for Land and Power in Liberia*. The New Press.

Moore, D. S., Pandian, A., & Kosek, J. (2003). 'The Cultural Politics of Race and Nature: Terrains of Power and Practice'. In D. Moore, J. Kosek, & A. Pandian (eds.), *Race, Nature, and the Politics of Difference* (1–70). Duke University Press.

Moore, J. (2010). 'The End of the Road? Agricultural Revolutions in the Capitalist World-Ecology, 1450–2010'. *Journal of Agrarian Change*, *10*(3), 389–413.

Moore, J. (2015). *Capitalism in the Web of Life: Ecology and the Accumulation of Capital*. Verso.

Moran-Thomas, A. (2019). *Traveling with Sugar: Chronicles of a Global Epidemic*. University of California Press.

Morgan, J. (2004). *Laboring Women: Reproduction and Gender in New World Slavery*. University of Pennsylvania Press.

Moulton, A., & Harris, D. eds. (2025). *Political Ecologies of Futurity: Storytelling Plantation Afterlives, Climate Erasures, and Socioecological Justice*. Bloomsbury.

p. 75 Münster, U. (2014). 'Working for the Forest: The Ambivalent Intimacies of Human–Elephant Collaboration in South Indian Wildlife Conservation'. *Ethnos*, *81*(3), 425–447.

Myers, N. (2017). 'From the Anthropocene to the Planthroposcene: Designing Gardens for Plant/People Involution'. *History and Anthropology*, *28*(3), 297–301.

Neimanis, A. (2023). 'Stygofaunal Worlds: Subterranean Estrangement and Otherwise Knowing for Multispecies Justice'. *Cultural Politics*, *19*(1), 18–38.

Nunn, E. (2016). '"A Great Addition to Their Harmony": Plantation Slavery and Musical Exchange in Seventeenth-Century Barbados'. *The Global South*, *10*(2), 27–47.

Obidzinski, K., Andriani, R., Komarudin, H., & Andrianto, A. (2012). 'Environmental and Social Impacts of Oil Palm Plantations and their Implications for Biofuel Production in Indonesia'. *Ecology and Society*, *17* (1), 25.

Oliveira, G., & Hecht, S. (2016). 'Sacred Groves, Sacrifice Zones, and Soy Production: Globalization, Intensification, and Neo-Nature in South America'. *The Journal of Peasant Studies*, *43*(2), 251–285.

Ortiz F. (1947). *Cuban Counterpoint: Tobacco and Sugar*. Translated by H. de Ortís. Duke University Press.

Ovstehage, A. (2021). 'Working the Plantationocene'. *Exertions*. 1–9.

Owens, D.C. (2017). *Medical Bondage: Race, Gender, and the Origins of American Gynecology*. University of Georgia Press.

Page, S. E., & Hooijer, A. (2016). 'In the Line of Fire: The Peatlands of Southeast Asia'. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences*, *371*(1696), 20150176.

Paredes, A. (2023). Experimental Science for the 'Bananapocalypse': Counter Politics in the Plantationocene. *Ethnos*, *88*(4), 837–863.

Pascoe, B. (2014). *Dark Emu*. Magabala Books.

Peluso, N. (1992). *Rich Forests, Poor People: Resource Control and Resistance in Java*. University of California Press.

Perdue, T. (2012). 'The Legacy of Indian Removal'. *The Journal of Southern History*, *78*(1), 3–36.

Perfecto, I., Jiménez-Soto, E., & Vandermeer, J. (2019). 'Coffee Landscapes: Shaping the Anthropocene. Forced Simplification on a Complex Agroecological Landscape'. *Current Anthropology*, *60*, S236–S250.

Perry, K.-K. Y. (2013). *Black Women against the Land Grab: The Fight for Racial Justice in Brazil*. University of Minnesota Press.

Petrenko, C., Paltseva, J., & Searle, S. (2016). *Ecological Impacts of Palm Oil Expansion in Indonesia*. The International Council on Clean Transportation.

p. 76

Plucknett, D. L. (1979). *Managing Pastures And Cattle Under Coconuts*. Routledge.

Plumwood, V. (2002). *Environmental Culture: The Ecological Crisis of Reason*. Routledge.

Porter, K. (2013). *The Black Seminoles: History of a Freedom-Seeking People*. University Press of Florida.

Price, C., & Chao, S. (2022). 'Multispecies, More-than-human, Non-human, Other-than-human: Reimagining Idioms of Animacy in an Age of Planetary Unmaking'. *Exchanges: The Interdisciplinary Research Journal*, *10*(2), 177–193.

Price, R. (1979). *Maroon Societies: Rebel Slave Communities in the America*. Johns Hopkins University Press.

Pye, O. Daud, R. Harmono, Y. & Tatat (2012). 'Precarious Lives: Transnational Biographies of Migrant Oil Palm Workers'. *Asia Pacific Viewpoint*, *53*(3), 330–342.

Qaim, M., Sibhatu, K. T., Siregar, H., & Grass, I. (2020). 'Environmental, Economic, and Social Consequences of the Oil Palm Boom'. *Annual Review of Resource Economics*, *12*, 321–344.

Ray, S. (2020). 'Indigo'. In *Indigo: Selected Stories*. Penguin Books, 85–96.

Rekow, L. (2019). 'Socio-ecological Implications of Soy in the Brazilian Cerrado'. *Challenges in Sustainability*, *7*(1), 7–29.

Roberts, J. (2019). '"We Live Like This": Local Inequalities and Disproportionate Risk in the Context of Extractive Development and Climate Change on New Hanover Island, Papua New Guinea'. *Oceania*, *89* (1), 68–88.

Rood, D. (2017). *The Reinvention of Atlantic Slavery: Technology, Labor, Race, and Capitalism in the Greater Caribbean*. Oxford University Press.

Rose, D. B., & van Dooren, T. (2017). 'Introduction: Telling Extinction Stories'. In T. van Dooren & M. Chrulew (eds.), *Extinction Studies: Stories of Time, Death, and Generations* (1–17). Columbia University Press.

Bird Rose, D., T. van Dooren, M. Chrulew, S. Cooke, M. Kearnes, and E. O'Gorman (2012). 'Thinking Through the Environment, Unsettling the Humanities'. *Environmental Humanities*, *1*, 1–5.

Rudge, A. (2022). 'Cultivating "Care": Colonial Botany and the Moral Lives of Oil Palm at the Twentieth Century's Turn'. *Comparative Studies in Society and History*, *64*(4), 878–909.

Rudge, A. (2024). 'Longing, Loss, and Regret Beyond the Borders of a Malaysian Oil Palm Plantation'. *Antipode Online*. 4 January. https://antipodeonline.org/wp-content/uploads/2023/12/3_Rudge.pdf.

p. 77

Rugemer, E. (2013). 'The Development of Mastery and Race in the Comprehensive Slave Codes of the Greater Caribbean during the Seventeenth Century'. *The William and Mary Quarterly*, *70*(3), 429–458.

Ruiz-Serna, D. (2023). 'Transitional Justice beyond the Human: Indigenous Cosmopolitics and the Realm of Law in Colombia'. *Cultural Politics*, *19*(1), 57–76.

Rusert, B. (2019). 'Naturalizing Coercion: The Tuskegee Experiments and the Laboratory of Life of the Plantation'. In R. Benjamin (ed.), *Captivating Technology: Race, Carceral Technoscience, and Liberatory Imagination in Everyday Life* (25–49). Duke University Press.

Sanadhya, T. (1991). 'The Story of the Haunted Line.' Translated by B. V. Lal and B. Shineberg. *The Journal of Pacific History*, *26*(1): 107–112.

Sarkar, A. (2023). 'The Fever Comes and Goes: Encountering Ecohorror, Toxicity, and Dystopian Impulses in Samanta Schweblin's Fever Dream'. *Literary Voice*, *1*(21), 176–184.

Sauer, S. (2018). 'Soy Expansion into the Agricultural Frontiers of the Brazilian Amazon: The Agribusiness Economy and Its Social and Environmental Conflicts'. *Land Use Policy*, *79*, 326–338.

Savilaakso, S., Garcia, C., Garcia-Ulloa, J. et al. (2014). *Systematic Review of Effects on Biodiversity from Oil Palm Production*. Centre for International Forestry Research.

Schneider, M., Goldman, L., Weisse, M., Amaral, L., & Calada, L. (2021). 'The Commodity Report: Soy Production's Impact on Forests in South America'. *Global Forest Watch*. December 3. www.globalforestwatch.org/blog/forest-insights/soy-production-forests-south-america/.

Scott, J. C. (1998). *Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed*. Yale University Press.

Sen, D. (2017). *Everyday Sustainability: Gender Justice and Fair Trade Tea in Darjeeling*. State University of New York Press.

Sharpe, C. (2010). *Monstrous Intimacies: Making Post-Slavery Subjects*. Duke University Press.

Sharpe, C. (2016). *In the Wake: On Blackness and Being*. Duke University Press.

Shiva, V. (1993). *Monocultures of the Mind: Perspectives on Biodiversity and Biotechnology*. Zed Books.

p. 77–78 Sklodowska, E. (2022). 'Etched in Sugar, Soil, Metal and Blood: The Plantationocene and the Afterlives of Racialized Plantation in Contemporary Cuban Art'. In M. Moraña (ed.), *Poetics of Race in Latin America* (35–60). Anthem Press.

p. 78 Soluri, J. (2000). 'People, Plants, and Pathogens: The Eco-Social Dynamics of Export Banana Production in Honduras, 1875–1950'. *Hispanic American Historical Review*, *80*(3), 463–502.

Soluri, J. (2005). *Banana Cultures: Agriculture, Consumption, and Environmental Change in Honduras and the United States*. University of Texas Press.

Song, X. P., Hansen, M. C., Potapov, P. et al. (2021). 'Massive Soybean Expansion in South America since 2000 and Implications for Conservation'. *Nature Sustainability*, *4*, 784–792.

Spiegel, M. (1996). *The Dreaded Comparison: Human and Animal Slavery*. Mirror Books.

Spillers, H. (1987). 'Mama's Baby, Papa's Maybe: An American Grammar Book'. *Diacritics*, *17*(2), 64–81.

Stevenson, E. & Buffavand, L. (2018). '"Do Our Bodies Know Their Ways?" Villagization, Food Insecurity, and Ill-Being in Ethiopia's Lower Omo Valley'. *African Studies Review*, *61*(1), 109–133.

Stoler, A. L. (1985). *Capitalism and Confrontation in Sumatra's Plantation Belt, 1879–1979*. Yale University Press.

Stone, G. and Glover, D. (2016). "Disembedding Grain: Golden Rice, the Green Revolution, and Heirloom Seeds in the Philippines." *Agriculture and Human Values*, *34*(1), 87–102.

Subramani. (1988). *The Fantasy Eaters: Stories from Fiji*. Three Continents Press.

Suhardi, E. (2025). 'Prabowo Reinvigorates the Development of Oil Palm Plantations.' *The Jakarta Post*. January 6. www.thejakartapost.com/opinion/2025/01/06/prabowo-reinvigorates-the-development-of-oil-palm-plantations.html.

Sultana, F. (2024). *Confronting Climate Coloniality: Decolonizing Pathways for Climate Justice*. Routledge.

Takaki, R. (1983). *Pau Hana: Plantation Life and Labor in Hawai'i, 1835–1920*. University of Hawai'i Press.

TallBear, K. (2015). 'An Indigenous Reflection on Working Beyond the Human/Not Human'. *GLQ: A Journal of Lesbian and Gay Studies*, *21*(2–3), 230–235.

Tammisto, T. (2019) "Making Temporal Environments: Work, Places, and History in the Mengen Landscape," in A. Lounela, E. Berglund, and T. Kallinessn (eds.) *Dwelling in Political Landscapes: Contemporary Anthropological Perspectives* (247–263). Suomalaisen Kirjallisuuden Seura.

p. 79 Tammisto, T. (2024). *Hard Work: Producing Places, Relations and Value on a Papua New Guinea Resource Frontier*. University of Helsinki Press.

Taussig, M. (2015). *The Corn Wolf*. University of Chicago Press.

Taussig, M. (2018). *Palma Africana*. University of Chicago Press.

Thomas, D. (2019). *Political Life in the Wake of the Plantation: Entanglement, Witnessing, Repair*. Duke University Press.

Thomas, D. (2023). 'Afterlives: The Recursive Plantation', in C. Le Petitcorps, M. Macedo, & I. Peano (eds.), *Global Plantations in the Modern World: Sovereignties, Ecologies, Afterlives* (353–364). Palgrave Macmillan.

Thompson, A. (2006). *Flight to Freedom: African Runaways and Maroons in the Americas*. University of the West Indies Press.

Thompson, T. (1975). *Plantation Societies, Race Relations, and the South: The Regimentation of Populations*. Duke University Press.

Tomich, D. (2011). 'Rethinking the Plantation: Concepts and Histories'. *Review (Fernand Braudel Center)*, *34*(1–2), 15–39.

Torres Mora, Á. G. (2020). 'Acumulación primitiva y acaparamiento de tierras en Colombia: Dinámicas del desarrollo'. *Papel Político*, *25*.

Trouillot, M.-R. (2002). 'Culture on the Edges: Caribbean Creolization in Historical Context'. In B. K. Axel (ed.), *From the Margins: Historical Anthropology and its Futures* (189–210). Duke University Press.

Tschakert, P., Schlosberg, D., Celermajer, D. et al. (2021). 'Multispecies Justice: Climate-Just Futures With, For, and Beyond Humans'. *Wiley Interdisciplinary Reviews*, *12*(2), e699.

Tsing, A. L. (2003). 'Natural Resources and Capitalist Frontiers'. *Economic and Political Weekly*, *38*(48), 5100–5106.

Tsing, A. L. (2011). 'Arts of Inclusion, or, How to Love a Mushroom'. *Australian Humanities Review*, *50*, 5–22.

Tsing, A. L. (2012). 'On Nonscalability: The Living World is Not Amenable to Precision-Nested Scales'. *Common Knowledge*, *18*(3), 505–524.

Tsing, A. L. (2014). 'More-Than-Human Sociality: A Call for Critical Description'. In K. Hastrup (ed.), *Anthropology and Nature* (27–42). Routledge.

Tsing, A. L. (2015). *The Mushroom at the End of the World: On the Possibility of Life in Capitalist Ruins*. Princeton University Press.

Tsing, A. L. (2018). 'Getting by in Terrifying Times'. *Dialogues in Human Geography*, *8*(1), 73–76.

Tsing, A. L., Mathews, A. S., & Bubandt, N. (2019). 'Patchy Anthropocene: Landscape Structure, Multispecies History, and the Retooling of Anthropology'. *Current Anthropology*, *60*, pS186–S197.

Union of Concerned Scientists. (2016). 'What's Driving Deforestation?' Union of Concerned Scientists. www.ucsusa.org/resources/whats-driving-deforestation.

p. 80

van Andel, T., Maat, H., & Pinas, N. (2023). 'Maroon Women in Suriname and French Guiana: Rice, Slavery, Memory'. *Slavery & Abolition*, *45*(2), 187–211.

van Dooren, T. (2014). *Flight Ways: Life and Loss at the Edge of Extinction*. Columbia University Press.

van Dooren, T., & Rose, D.B. (2016). 'Lively Ethography: Storying Animist Worlds'. *Environmental Humanities*, *8*(1), 77–94.

van Dooren, T., Kirksey, E. S., & Münster, U. (2016). 'Multispecies Studies: Cultivating Arts of Attentiveness'. *Environmental Humanities*, *8*(1), 1–23.

Vandertop, C. (2020). 'Ghosts of the Plantation: Sugar, Narrative Energetics and Gothic Ecologies in Fiji'. *Green Letters*, *24*(2), 155–168.

Watkins, C. (2021). *Palm Oil Diaspora: Afro-Brazilian Landscapes and Economies on Bahia's Dendê Coast*. Cambridge University Press.

Webb, M., & Webb-Gannon, C. (2020). 'Decolonization, Popular Song and Black-Pacific Identity in Melanesia'. *Media, Culture & Society*, *42*(1), 142–151.

Webb-Gannon, C., & Webb, M. (2018). 'Sound Tracks of the Black Pacific: Music, Identity and Resilience in Australian South Sea Islander Communities'. *Pacific Dynamics: Journal of Interdisciplinary Research*, *2* (1), 35–44.

Weheliye, A. G. (2014). *Habeas Viscus: Racializing Assemblages, Biopolitics, and Black Feminist Theories of the Human*. Duke University Press.

West, P. (2012). *From Modern Production to Imagined Primitive: The Social World of Coffee from Papua New Guinea*. Duke University Press.

Whitaker, J.A. (2020). 'Strategic Alliance and the Plantationocene among the Makushi in Guyana.' *Social Anthropology*, *28*, 881–896.

White, M. (2018). *Freedom Farmers: Agricultural Resistance and the Black Freedom Movement*. University of North Carolina Press.

Williams, B., & Porter, J. (2022). 'Cotton, Whiteness, and Other Poisons'. *Environmental Humanities*, *14*(3), 499–521.

Williams, B., Squire, D., & Tuitt, F. eds. (2021). *Plantation Politics and Campus Rebellions: Power, Diversity, and the Emancipatory Struggle in Higher Education*. State University of New York Press.

Winter, C. (2023). 'Unearthing the Time/Space/Matter of Multispecies Justice'. *Cultural Politics*, *19*(1), 39–56.

Wolf, E. R., & Mintz, S. W. (1957). 'Haciendas and Plantations in Middle America and the Antilles'. *Social and Economic Studies*, *6*(3), 380–412.

p. 81 Wolford, W. (2021). 'The Plantationocene: A Lusotropical Contribution to the Theory'. *Annals of the American Association of Geographers*, *111*(6), 1622–1639.

Woods, C. (1998). *Development Arrested: The Blues and Plantation Power in the Mississippi Delta*. Verso Books.

World Resources Institute. (2024). 'Deforestation Linked to Agriculture Indicator'. Global Forest Review, updated April 4, 2024. World Resources Institute. https://research.wri.org/gfr/forest-extent-indicators/deforestation-agriculture.

Wynter, S. (1971). "Novel and History, Plot and Plantation," *Savacou*, *5*, 95–102.

Wynter, S. (2003). 'Unsettling the Coloniality of Being/Power/Truth/Freedom: Towards the Human, AfterMan, Its Overrepresentation—An Argument'. *CR: The Centennial Review*, *3*(3), 257–337.

Yates-Doerr, E. (2019). 'Whose Global, Which Health? Unsettling Collaboration with Careful Equivocation'. *American Anthropologist*, *121* (2), 297–310.

Acknowledgements

Research towards this publication was supported by the Wenner-Gren Foundation for Anthropological Research (Gr. 9942, EAG-136, Gr. 9196) and the Australian Research Council (DE220100025).

For Jacob, and all the worlds you make possible.

PDF 88

Environmental Humanities

PDF 89

Serenella Iovino *University of North Carolina at Chapel Hill*

Serenella Iovino is Professor of Italian Studies and Environmental Humanities at the University of North Carolina at Chapel Hill. She has written on a wide range of topics, including environmental ethics and ecocritical theory, bioregionalism and landscape studies, ecofeminism and posthumanism, comparative literature, eco-art, and the Anthropocene.

Timo Maran *University of Tartu*

Timo Maran is an Estonian semiotician and poet. Maran is Professor of Ecosemiotics and Environmental Humanities and Head of the Department of Semiotics at the University of Tartu. His research interests are semiotic relations of nature and culture, Estonian nature writing, zoosemiotics and species conservation, and semiotics of biological mimicry.

Thom van Dooren *The University of Sydney*

Thom van Dooren is Professor of Environmental Humanities at the School of Humanities and the Sydney Environment Institute at the University of Sydney. His research focuses on processes of biocultural unravelling and resurgence at the edge of extinction, drawing on the literatures and approaches of philosophy, anthropology, cultural studies, and feminist science studies.

About the Series

The environmental humanities is a new transdisciplinary complex of approaches to the embeddedness of human life and culture in all the dynamics that characterize the life of the planet. These approaches reexamine our species' history in light of the intensifying awareness of drastic climate change and ongoing mass extinction. To engage this reality, Cambridge Elements in Environmental Humanities builds on the idea of a more hybrid and participatory mode of research and debate, connecting critical and creative fields.

Environmental Humanities Elements in the Series

PDF 90

Aging Earth: Senescent Environmentalism for Dystopian Futures Jacob Jewusiak

Blue Humanities: Storied Waterscapes in the Anthropocene Serpil Oppermann

Nonhuman Subjects: An Ecology of Earth-Beings Federico Luisetti

Indigenous Knowledge and Material Histories: The Example of Rubber Jens Soentgen

The Open Veins of Modernity: Ecological Crisis and the Legacy of Byzantium and Pre-Columbian America Eleni Kefala

Slime: An Elemental Imaginary Simon C. Estok

Growing Hope: Narratives of Food Justice Alexa Weik von Mossner

Descartes and the Non-Human Emma Gilby

Automobility and the Anthropocene: The Car as Post-Human Gordon M. Sayre

The Earth Intoxicated on Imagination Annabelle Dufourcq

Ruin Ecology: An Exercise in Environmental Imagination Marco Malvestio

Plantations: Extraction, Extinction, Emergence Sophie Chao

A full series listing is available at: www.cambridge.org/EIEH